

When Does Ability Grouping Improve Learning?

Evidence from Two Experiments in Kenya and Liberia*

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Abstract

In many school systems, students in adjacent grades overlap substantially in actual learning levels. One response is to reassign students across adjacent grades into instructional tracks based on a diagnostic assessment. Such regrouping may improve instructional targeting, but it also changes class size, peer composition, and age-grade structure. We study two randomized evaluations of cross-grade reading groups in Kenya and Liberia. Neither intervention generates evidence of average achievement gains, and the diagnostics point to different constraints. In Liberia, the baseline assessment explains less than 7% of endline variation, and regrouping generates little endline-predictive sorting improvement while increasing class size by roughly one-third. In Kenya, the baseline explains more than 50% of endline variation and regrouping substantially compresses within class baseline score dispersion, yet there is no evidence of average score gains. Because assigned lesson content is fixed by track under scripted curricula, the Kenya experiment rules out a simple information or compliance failure: the implemented rule changes measured sorting and classroom composition, but movers and students with larger predicted assignment gains do not benefit more. Approximate peer/rank specifications point to adverse classroom-composition changes, while exact control function and recentering checks show that this evidence should not be interpreted as an identified causal peer/rank coefficient. The broader result is that assignment rules face two hurdles: the diagnostic must contain useful information, and the classrooms created by acting on it must convert improved measured placement into productive learning environments.

Keywords: ability grouping; tracking; diagnostic assessment; peer effects; class size; education in developing countries.

JEL codes: I21; I24; I28; O15.

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1 Introduction

Schools must assign heterogeneous students to a limited number of instructional environments. The standard rule is grade, which groups students by age and prior progression through the curriculum. Grade is simple to administer, and it carries useful information about curricular exposure and developmental stage. Yet grade can be poorly aligned with actual ability when students within the same grade differ sharply in achievement and students in adjacent grades overlap substantially in skill.

This is a particular instance of a broader economic problem: how should an organization assign heterogeneous individuals to production environments when the available signals are imperfect? Schools assign students to classrooms, firms assign workers to tasks and teams, hospitals assign patients to units, and lenders assign borrowers to contracts. In each case, a new diagnostic or screening tool may improve assignment by moving beyond a coarse incumbent rule. But acting on that signal also changes the organization itself. It reshapes who works with whom, who ranks where within the group, and how congested or manageable each production unit becomes. The value of a better signal therefore cannot be judged by predictive power alone. It depends on whether the gains from improved assignment outweigh the organizational costs created by using the signal.

These tradeoffs are especially salient in many developing-country school systems. Achievement variance within grades is wide, automatic promotion policies leave many students far below grade-level expectations, and multi-grade classrooms are common in small or under-resourced schools (Banerjee et al., 2007; Muralidharan et al., 2019). In such settings, a single teacher may face a classroom in which some students cannot recognize letters while others read fluently. Cross-grade ability grouping is an appealing response. It requires no new teachers, materials, or technology, only a reassignment of existing students to existing classrooms based on a diagnostic assessment. And in schools with a single class per grade, within grade tracking is often not feasible: there are too few students within a grade to form multiple instructional groups without pooling across grades. Cross-grade grouping is therefore the natural organizational form when schools want to teach at assessed level rather than enrolled grade. Recent evidence that aligning instruction to students' demonstrated level can improve learning reinforces the case for trying it (Duflo et al., 2020, 2011).

But cross-grade grouping is also a demanding form of assignment. Relative to within grade

tracking, it does more than change the instructional target. It reorganizes classrooms along several margins at once. It changes peer composition, alters students' relative standing within the classroom, can increase or decrease class size, and pools students from different grades into the same instructional space (Betts, 2011). A reassignment rule that looks superior on paper may therefore fail in practice if the classrooms it creates are less productive than the ones it replaces.

We study this question using two randomized evaluations of cross-grade reading groups in schools operated or managed by NewGlobe Education, an education provider that uses scripted lesson plans delivered via digital tablets. In 2018, school grade group units in 54 Liberian public schools managed by NewGlobe were randomized to regroup students in Grades 1–4 into two reading tracks based on a baseline reading assessment. In the same year, 50 of 292 NewGlobe schools in Kenya were randomized to regroup students in Grades 1–2 into two reading groups based on a combined literacy and English test. In both cases, regrouping applied to reading instruction only: students were reassigned to cross-grade reading classrooms for a daily instructional block, then returned to their enrolled grade for all other subjects. Because NewGlobe uses scripted curricula, instructional level in each track was determined by the track designation rather than by teacher discretion.

Neither experiment provides evidence of average learning gains. The Liberia estimate is negative (-0.212 SD, $p = 0.12$), while the Kenya estimate is small, positive, and statistically indistinguishable from zero (0.031 SD, $p = 0.56$). The adjusted Kenya specification is precise enough that its 95% upper confidence bound lies below the 0.18 SD gain found by Duflo et al. (2011) for within grade tracking. Together with the assignment and classroom diagnostics, the two experiments do more than report two imprecise average effects. They point to two requirements for score-based grouping to work: the diagnostic must improve assignment relative to grade, and the resulting classrooms must convert that improved assignment into learning.

In Liberia, the diagnostics point first to the diagnostic hurdle. The baseline assessment is a very weak predictor of endline achievement, explaining less than 7% of endline variation. Regrouping based on it does reduce measured EB-to-track-target mismatch, but it barely compresses within class baseline score dispersion—the sorted classrooms remain nearly as heterogeneous as the unsorted ones on the diagnostic margin. Meanwhile, regrouping increases class sizes by roughly 10 students. The negative point estimates are concentrated in the upper track (-0.348 SD, $p = 0.03$)

and extend across the ability distribution rather than clustering near the cutoff—a pattern more consistent with constraints in the upper-track classroom than latent misassignment near the threshold. Observed treated-student assignments follow the published score cutoffs with zero observed assignment error; the measured movement toward implemented track targets is not accompanied by positive achievement effects when the same rule also reorganizes classrooms along class-size, grade-mixing, and peer-composition margins.

Kenya clears the signal-quality part of this test but exposes the second requirement. The baseline is far more predictive, explaining over 50% of endline variation, and regrouping does compress within class baseline score dispersion. Yet there is no evidence of average achievement gains. Scripted curricula help interpret this puzzle because assigned lesson content is fixed by track: two students with the same baseline score, grade, and track are assigned the same scripted lesson sequence. The assignment-payoff diagnostics sharpen the result. Students whose reading group changes under treatment do not benefit more than stayers, treatment effects do not rise with distance from the relevant cutoff among movers, and a predicted assignment-gain index has a near-zero interaction with treatment. The peer/rank evidence is diagnostic rather than definitive: approximate specifications are negative and large enough to matter in accounting terms, while exact control function and recentered specifications are close to zero. We therefore do not claim to isolate a single offsetting peer/rank coefficient. The Kenya lesson is broader: the rule successfully changed measured sorting and classroom composition, but the assignment margin it created did not translate the sharper diagnostic into average learning gains.

A simple model of score-based grouping organizes these findings. It decomposes the effect of replacing grade-based with score-based assignment into four channels—assignment gains from reducing instructional mismatch, changes in productive instructional time, peer-composition effects, and relative-rank effects—and delivers a benchmark threshold condition: even when the score is used to minimize expected mismatch, regrouping raises achievement only if the informational gain from the diagnostic exceeds the costs of classroom reorganization. The experiments then evaluate the realized, pre-specified cutoff rules implemented by NewGlobe.

The paper contributes to several literatures. We provide randomized evidence on cross-grade ability grouping—what the early American literature called the Joplin Plan (Floyd, 1954).¹ Meta

¹There is also a large body of research on between-school ability grouping, mainly from Europe where tracking

analyses of within grade tracking generally find small or null average effects (Slavin, 1987, 1990; Kulik and Kulik, 1987; Terrin and Triventi, 2022), with heterogeneous results across tracks (Burke and Sass, 2013; Hoxby and Weingarth, 2005; Antecol et al., 2016; Garlick, 2018; de Roux and Riehl, 2022). Slavin (1987) judged the early Joplin Plan studies promising, but they were small and none used random assignment. In developing countries, the benchmark is Duflo et al. (2011), who randomly assigned Kenyan primary schools to within grade tracking and found average gains of 0.18 SD. Cross-grade grouping is a harder problem: it pools students across grades, requires comensurable rankings across distributions that may be shrunk at different rates, and mechanically increases grade mixing and class size variation.

The two-experiment design gives us leverage on why similar null average effects can arise from different constraints, not only on whether ability grouping raises achievement. The tracking literature documents substantial heterogeneity across contexts (Betts, 2011; Card and Giuliano, 2016; Vardardottir, 2013), but the sources are difficult to isolate because regrouping simultaneously changes instructional targeting, peer composition, class size, and relative standing. Our two experiments share an operator, scripted reading curricula, deterministic cutoff rules, and randomized assignment, but differ sharply in signal quality and classroom reallocation. In Kenya, scripted curricula fix assigned lesson content by track, so conditioning on baseline score bin, treatment status, and grade approximately absorbs the instructional-targeting channel and lets us study the social consequences of regrouping with less direct confounding from instructional targeting. This connects to work on how peer effects depend on group structure (Hoxby and Weingarth, 2005; Rjosk, 2022; Vigdor and Nechyba, 2007) and on how assignment rules change outcomes by altering interaction patterns within the assigned group (Carrell et al., 2013; Garlick, 2018).

The peer effects and rank literatures motivate two of the channels in our framework. Higher-achieving peers may generate positive learning spillovers, but regrouping also changes a student’s ordinal rank, which affects achievement independently (Murphy and Weinhardt, 2020; Bertoni and Nisticò, 2023). Since tracking moves both margins at once—typically raising peer quality while lowering rank for students sorted upward—the net social effect is not signed *a priori*.

Finally, we develop a framework for evaluating assignment rules that accounts for informational

between academic and vocational tracks is prevalent (Fu and Mehta, 2018; Lavrijsen and Nicaise, 2015). We focus on within-school ability grouping.

precision and the organizational consequences of acting on it. The Teaching at the Right Level literature asks whether instruction targeted to assessed level raises scores (Banerjee et al., 2017; Duflo et al., 2020); recent work emphasizes implementation fidelity and subgrouping precision (Angrist and Meager, 2023). Our framework addresses a different question: whether a score-based rule improves on the incumbent rule—grade—once the resulting changes in classroom organization are taken into account. Within-grade tracking in Kenya raised achievement (Duflo et al., 2011); ability grouping in Mexico generated gains concentrated among initially higher-achieving students (Busso and Frisncho, 2023). Our model treats assignment gains, peer composition, relative rank, and productive instructional time as joint components of the treatment, so the value of a diagnostic depends on both its information content and the productivity of the classrooms created by using it.

The paper proceeds as follows. Section 2 develops the theoretical framework. Section 3 describes the experiments and data. Section 4 presents the empirical strategy. Section 5 reports results. Section 6 discusses robustness. Section 7 concludes.

2 Theoretical Framework

We model cross-grade ability grouping as a classification problem under a noisy signal. Students differ in latent ability, but schools observe only an imperfect diagnostic. A score-based reassignment rule can improve achievement by placing students in classrooms better matched to their level, but it also reorganizes the classroom: changing class sizes, mixing students from different developmental stages, shifting peer composition, and altering students’ relative standing.

2.1 Environment and Assignment

Student i in grade $g(i) \in \mathcal{G}$ has latent ability

$$\theta_i = \mu_{g(i)} + \nu_i, \tag{1}$$

where $\mu_{g(i)}$ is the grade-specific mean and ν_i captures within grade heterogeneity, with $\mathbb{E}[\nu_i] = 0$ and $\text{Var}(\nu_i \mid g(i)) = \sigma_{\nu,g}^2$. The school does not observe θ_i directly. It observes a noisy diagnostic

score

$$s_i = \theta_i + u_i, \quad (2)$$

with $\mathbb{E}[u_i \mid \theta_i, g(i)] = 0$ and $\text{Var}(u_i \mid g(i)) = \sigma_{u,g}^2$. The school’s problem is analogous to an employer deciding whether to use a test score in addition to an observable credential (here, grade) when assigning workers to tasks (Autor and Scarborough, 2008). Grade already provides some information about ability; the diagnostic is valuable only if it moves the school meaningfully beyond what grade already reveals. When the test is noisy relative to the within grade ability distribution, the posterior $\mathbb{E}[\theta_i \mid s_i, g(i)]$ is pulled back toward the grade mean, and reassignment based on the score is little better than leaving students in their enrolled grade.

The posterior expectation of ability given the signal and grade is

$$a_i \equiv \mathbb{E}[\theta_i \mid s_i, g(i)] = \mu_{g(i)} + \rho_g(s_i - \mu_{g(i)}), \quad (3)$$

where

$$\rho_g \equiv \frac{\sigma_{\nu,g}^2}{\sigma_{\nu,g}^2 + \sigma_{u,g}^2} \quad (4)$$

is the signal quality of the diagnostic. When ρ_g is high, the posterior tracks the raw score closely and the diagnostic carries substantial information beyond grade; when ρ_g is low, the posterior is pulled toward the grade mean and reassignment based on the test is only a weak improvement over grade assignment. The parameter ρ_g is not directly observed. Our empirical proxy is R_g^2 , the within grade R^2 from regressing endline scores on baseline scores in the control group. Under the model’s Gaussian structure, $R_g^2 = \rho_g \cdot \sigma_{\nu,g}^2 / (\sigma_{\nu,g}^2 + \sigma_{\varepsilon,g}^2)$, where $\sigma_{\varepsilon,g}^2$ is endline noise; because endline noise attenuates the correlation, $R_g^2 < \rho_g$, so the empirical measure is a conservative proxy for the model’s shrinkage weight. We refer to R_g^2 as *baseline predictive power*. It blends test precision, achievement persistence, and baseline-endline construct alignment into a single measure of how useful the diagnostic is for assignment—whatever the source of a low R_g^2 , a signal with low predictive power provides less useful information for reassignment.

Under the status quo regime (G), students remain in grade-based classrooms and receive grade-level instruction $I_i^G = I(g(i))$. Under score-based grouping (T), the school assigns students

to a low or high instructional track using the posterior signal:

$$k_i^T = \begin{cases} L & \text{if } a_i < c, \\ H & \text{if } a_i \geq c, \end{cases} \quad I_i^T = \begin{cases} I_L & \text{if } k_i^T = L, \\ I_H & \text{if } k_i^T = H. \end{cases} \quad (5)$$

Under scripted curricula—where lesson plans are standardized and teacher discretion over content is limited—assigned instructional level is fixed by track designation rather than deliberately retargeted by the teacher. This means that assigned lesson content depends primarily on which track a student enters rather than being deliberately retargeted to the realized composition of her classroom. Both experiments we study operate under scripted instruction (Section 3). The assignment decision—high or low track—is therefore predetermined by the score rule, while the classroom-reorganization consequences—peer quality, class size, and grade mixing—are realized through the composition of the resulting classroom. This structure helps distinguish instructional-targeting from classroom composition channels in the empirical analysis, though both are induced by the same assignment rule.

2.2 Achievement

Endline achievement under assignment regime $a \in \{G, T\}$ is

$$Y_i(a) = \theta_i + \tau_{c(i,a)} \left[\alpha - \lambda(\theta_i - I_i(a))^2 + \pi \bar{\theta}_{-i,c(i,a)} + \eta r_{i,c(i,a)} \right] + \varepsilon_i(a). \quad (6)$$

Baseline ability θ_i persists into the endline: a student who learns nothing during the intervention still retains her initial achievement level. The bracketed term is the value added by instruction, which depends on how well instruction is matched to the student’s level, who her classmates are, and how much classroom time is productively used. This term is scaled by τ_c , so all instructional channels are attenuated when the classroom becomes harder to manage.

The term $(\theta_i - I_i(a))^2$ is *instructional mismatch*: a student whose ability is far from the instructional target of her classroom learns less than one whose ability is close to it, regardless of whether the instruction is too advanced or too basic. The quadratic form implies that the cost of mismatch is convex—small deviations matter little, but large deviations are increasingly costly.

Under grade-based assignment, mismatch arises because students within the same grade differ in ability but receive the same instruction. Under score-based grouping, the diagnostic can reduce mismatch by sorting students into tracks whose instructional targets are closer to their ability, but only if the diagnostic is informative enough for the sorting to be accurate.

The term $\bar{\theta}_{-i,c(i,a)}$ captures the level of peer ability in student i 's classroom, while $r_{i,c(i,a)} \equiv \text{rank}_i/N_{c(i,a)}$ denotes student i 's percentile rank within that classroom. We normalize by class size so that rank is comparable across classrooms of different sizes—an important consideration here, since tracking changes class sizes. Both terms are needed because regrouping can place a student among stronger peers while simultaneously lowering that student's relative standing, so the net social effect of tracking is not signed *a priori*. The sign of π is theoretically ambiguous—positive under peer learning, negative if lower-ability peers generate disruption. The coefficient η captures relative-standing effects: percentile rank affects achievement independently of the level of surrounding peers (Murphy and Weinhardt, 2020; Elsner et al., 2021), consistent with the “big-fish-little-pond” effect in educational psychology (Hoxby and Weingarth, 2005).

The object $\tau_{c(i,a)} \in (0, 1]$ is the fraction of classroom time that is productively engaged. Following the logic of Lazear (2001), productive time is a classroom-level public good: when the classroom becomes harder to manage, all students lose instructional time. We let

$$\tau_c = \tau(N_c, V_c^g), \quad \frac{\partial \tau}{\partial N_c} < 0, \quad \frac{\partial \tau}{\partial V_c^g} < 0, \quad (7)$$

where N_c is class size and $V_c^g \equiv \text{Var}_{i \in c}(g(i))$ is within classroom grade dispersion. The class size channel is standard: larger classrooms reduce the share of time available per student. The grade dispersion channel is specific to cross-grade grouping. A diagnostic sorts students on one measured dimension (reading level), but students from different grades differ on unmeasured dimensions—behavioral maturity, attention span, familiarity with classroom routines—that affect how easily the classroom functions as a unit. When regrouping pools adjacent grades, greater distance between students' developmental stages and the classroom's modal stage can raise the teacher's coordination and classroom-management burden. The model captures this as a productive-time cost that increases with within class grade dispersion, which rises mechanically in treatment classrooms.

Because τ_c multiplies the entire bracketed term, a fall in productive time attenuates all other

channels: if regrouping creates a classroom that is too large or too heterogeneous in developmental stage for the teacher to manage, the returns to better instructional match, stronger peers, or higher rank are all diminished. Correct placement does not help much in a classroom where effective instruction is not occurring.

2.3 Treatment Effect Decomposition

Define student i 's *assignment gain* as

$$B_i \equiv (\theta_i - I_i^G)^2 - (\theta_i - I_i^T)^2. \quad (8)$$

A positive B_i means that score-based grouping moves student i closer to the instructional level that fits her ability. Let

$$A_i(a) \equiv \alpha - \lambda(\theta_i - I_i(a))^2 + \pi \bar{\theta}_{-i,c(i,a)} + \eta r_{i,c(i,a)} \quad (9)$$

collect the within-brackets terms. The treatment effect for student i separates into two parts: changes in the quality of instruction holding productive time fixed, and changes in productive time holding instruction quality at its treatment level:

$$\Delta Y_i \equiv Y_i(T) - Y_i(G) = \tau_{c(i,G)} [A_i(T) - A_i(G)] + [\tau_{c(i,T)} - \tau_{c(i,G)}] A_i(T) + \tilde{\varepsilon}_i, \quad (10)$$

where $\tilde{\varepsilon}_i = \varepsilon_i(T) - \varepsilon_i(G)$. Expanding gives

$$\Delta Y_i = \underbrace{\lambda \tau_{c(i,G)} B_i}_{\text{assignment gain}} + \underbrace{\pi \tau_{c(i,G)} \Delta \bar{\theta}_i}_{\text{peer level}} + \underbrace{\eta \tau_{c(i,G)} \Delta r_i}_{\text{rank}} + \underbrace{[\tau_{c(i,T)} - \tau_{c(i,G)}] A_i(T)}_{\text{productive time}} + \tilde{\varepsilon}_i, \quad (11)$$

where $\Delta \bar{\theta}_i \equiv \bar{\theta}_{-i,c(i,T)} - \bar{\theta}_{-i,c(i,G)}$ and $\Delta r_i \equiv r_{i,c(i,T)} - r_{i,c(i,G)}$.

Score-based grouping therefore affects achievement through four channels: it can improve instructional match (B_i), change peer composition ($\Delta \bar{\theta}_i$), change relative rank (Δr_i), and alter the amount of productive instructional time available ($\Delta \tau_c$). Averaging over students yields

$$\text{ATE} = \lambda \mathbb{E}[\tau_{c(i,G)} B_i] + \pi \mathbb{E}[\tau_{c(i,G)} \Delta \bar{\theta}_i] + \eta \mathbb{E}[\tau_{c(i,G)} \Delta r_i] + \mathbb{E}[(\tau_{c(i,T)} - \tau_{c(i,G)}) A_i(T)]. \quad (12)$$

2.4 Signal Quality and the Value of Grouping

Under the Gaussian structure, the posterior a_i concentrates around θ_i as signal quality rises, so a school with access to a more informative posterior can make more accurate assignments. Because grade means can differ, a zero-signal rule may still reproduce a grade-only reassignment: when $\rho_g = 0$, all students in grade g receive the same posterior μ_g . The clean comparative-static object is therefore the *incremental* assignment gain relative to that zero-signal rule, $\Delta B_i(\rho) \equiv B_i(\rho) - B_i(0)$, where ρ indexes a nested information experiment that ranges from no within grade information to perfect observation of ability.

Proposition 1. *Under Gaussian ν_i and u_i , consider a nested information experiment indexed by $\rho \in [0, 1]$ and a two-track rule chosen to minimize expected mismatch given the available posterior information:*

- (i) *The expected incremental assignment gain $\mathbb{E}[\Delta B_i(\rho)]$ is weakly increasing in ρ . It satisfies*

$$\mathbb{E}[\Delta B_i(0)] = 0, \quad \mathbb{E}[\Delta B_i(1)] = \mathbb{E}[(\theta_i - I_i^{T0})^2 - (\theta_i - I_i^{T*})^2],$$

where I_i^{T0} is the track target assigned by the zero-signal rule and I_i^{T} is the oracle-optimal track target under perfect observation of θ_i . The increase is strict when additional signal precision changes the optimal track assignment with positive probability and the two track targets differ.*

- (ii) *If $ATE(\rho = 1) > 0$, $ATE(\rho)$ is continuous in ρ , and any zero-signal assignment gain is not large enough to offset the zero-signal reorganization terms so that $ATE(0) \leq 0$, then there exists at least one $\rho^* \in [0, 1)$ such that $ATE(\rho^*) = 0$. If the zero-signal condition is strict, $ATE(0) < 0$, the crossing is interior: $\rho^* \in (0, 1)$.*

Proof sketch. Part (i): When $\rho = 0$, the posterior $a_i = \mu_{g(i)}$ for all students in the same grade: the test provides no within grade information, so track assignment is determined entirely by grade means. This defines the zero-signal target I_i^{T0} and normalizes the incremental gain to $\Delta B_i(0) = 0$. As ρ rises in a nested information experiment, the posterior becomes more informative about θ_i in the Blackwell sense. A rule that observes the more informative posterior can ignore the extra

information and reproduce the assignment chosen under the less informative posterior, so the minimized expected mismatch cannot increase; it falls strictly when the additional signal changes the optimal assignment for a positive-mass set of students. At $\rho = 1$, the posterior equals true ability and assignment is oracle-optimal given the available tracks, yielding the maximal incremental mismatch reduction relative to I_i^{T0} . This establishes the unweighted value-of-information benchmark. The productive-time-weighted assignment term in equation (12) inherits the same monotonicity only under additional restrictions on the status quo weights; part (ii) therefore states the crossing condition directly in terms of the full ATE. Part (ii): At $\rho = 0$, the diagnostic contains no within grade information. If the reorganization terms induced by such grade-only regrouping are weakly nonpositive and any grade-only assignment gain is not enough to offset them, then $\text{ATE}(0) \leq 0$; combined with $\text{ATE}(1) > 0$ and continuity, the intermediate value theorem gives a zero at $\rho = 0$ when $\text{ATE}(0) = 0$ and an interior crossing when $\text{ATE}(0) < 0$. $\square$

The reorganization terms— $\mathbb{E}[\tau_{c(i,G)}\Delta\bar{\theta}_i]$, $\mathbb{E}[\tau_{c(i,G)}\Delta r_i]$, and $\mathbb{E}[(\tau_T - \tau_G)A_i(T)]$ —are themselves functions of signal quality through the assignment, so the threshold need not be unique. The proposition describes an optimal value-of-information benchmark. For an implemented cross-grade cutoff rule, poor signal quality can make the realized assignment term small or even negative relative to grade assignment, because the rule may pool students across grades on a weakly informative score. At the same time, near-random reassignment scrambles peer composition and—in settings where regrouping increases class sizes or grade mixing—reduces productive time. Thus the realized terms in equation (12) need not have the signs implied by the optimal assignment benchmark. The production function parameters $(\lambda, \pi, \eta, \tau_c)$ need not be the same across settings. The model asks whether the assignment gain outweighs the reorganization costs; the answer can differ across settings because signal quality, class sizes, and composition shifts differ.

The proposition is a benchmark for the value of information under the best two-track use of a nested posterior signal. It is not a claim that every implemented cutoff rule, or every vector of grade-specific shrinkage weights, produces monotone realized gains. The experiments below do not estimate or implement the optimal rule; they evaluate pre-specified cutoff rules chosen by the operator before outcomes were observed. The empirical question is therefore whether those realized rules generated enough assignment gain, given the classroom reorganization they induced, to raise achievement.

Remark 1 (Within-grade vs. cross-grade tracking). *The decomposition in equation (12) applies to any score-based reassignment, but the magnitudes of the terms differ systematically between within grade tracking (splitting a single grade into tracks) and cross-grade tracking (pooling adjacent grades and re-sorting).*

Productive time. Under within grade tracking, all students in a classroom share the same grade, so $V_c^g = 0$ by construction. Under cross-grade tracking, treatment classrooms pool students from different grades, so $V_c^g > 0$. Since $\partial\tau/\partial V_c^g < 0$, cross-grade tracking incurs a productive-time cost that within grade tracking avoids entirely. Class sizes also differ: within grade tracking splits an existing grade-level classroom roughly in half ($N_c^{WG} \approx N_g/2$), while cross-grade tracking determines class sizes by the ability distribution across the pooled grades, which can substantially increase N_c .

Assignment gain. Under within grade tracking with a grade-specific cutoff c_g , student i 's assignment depends on $a_i - c_g = \rho_{g(i)}(s_i - \mu_{g(i)}) - (c_g - \mu_{g(i)})$, which involves only the signal quality $\rho_{g(i)}$ of her own grade. Under a pooled cross-grade cutoff c , student i 's assignment depends on

$$a_i - c = \mu_{g(i)} + \rho_{g(i)}(s_i - \mu_{g(i)}) - c. \quad (13)$$

Whether student i in grade g is assigned above or below a student j in grade g' now depends on both ρ_g and $\rho_{g'}$. When signal quality varies across grades, posteriors are shrunk at different rates: a student in a high- ρ grade is ranked largely on ability, while a student in a low- ρ grade is ranked largely on her grade mean. The same issue appears with grade-specific cutoffs whenever the rule must compare or pool scores across grades. Cross-grade ranking can therefore conflate differently informative signals, distorting assignment in a way that within grade tracking does not face.

3 Experimental Design and Data

3.1 Setting

Both experiments take place in schools operated or managed by NewGlobe Education. In Kenya, NewGlobe operated a private-school network; in Liberia, it managed public schools under the Liberia Education Advancement Program. Across both settings, the model used scripted lesson plans delivered through digital tablets, with teacher compliance monitored by school managers.

Because lesson content is scripted, assigned instructional level is determined by track designation rather than by teacher discretion—a feature that helps separate instructional targeting from the peer/rank diagnostics in Section 4.3.

3.2 Experiment 1: Kenya (2018)

In August 2018, 50 of 292 NewGlobe schools in the full Kenya randomization roster were assigned to receive the reading group intervention. The analytic sample contains eligible students from 263 schools, including 46 treated schools. In treated schools, students in Grades 1 and 2 were assigned to one of two reading groups (“Blue” and “Purple”) based on a combined literacy and English baseline test administered in June 2018. Grade 1 students scoring above 40 points were assigned to the Purple (upper) group; Grade 2 students above 35 points.² Treatment was stratified by constituency. Students received 90 minutes of instruction with their reading group daily for Term 3 (approximately 3 months).

3.3 Experiment 2: Liberia (2018)

In Liberia, the experiment randomized school grade group units in 54 public schools managed by NewGlobe under the Liberia Education Advancement Program. Grades 1 and 2 formed one grade group and Grades 3 and 4 formed another, yielding 99 school grade group units in the analytic sample. In treated units, students were regrouped within the grade group and split into two tracks. The diagnostic test was a baseline reading assessment; the cutoff for the upper track was 23 points for the Grade 1–2 pool and 14 points for the Grade 3–4 pool. Treatment was stratified by school cohort and grade group.

3.4 Data

Administrative data were provided by NewGlobe for both experiments. In Kenya, the baseline instrument is a combined literacy and English midterm examination administered in Term 2 (June 2018), the same test used for track assignment. Historical scores in mathematics, language, and Kiswahili are available for a subset of students and serve as additional pre-treatment covariates.

²Cutoffs were chosen by NewGlobe before the experiment to approximate an equal split of each grade’s baseline score distribution, targeting roughly half of students in each track. The grade-specific thresholds reflect differences in the raw-score distributions across grades.

Administrative records also cover lesson completion rates and teacher attendance, which we use at the school-grade lesson-cell level to examine one observable component of the productive-time channel in Section 5. Schools in the Kenya sample had been operating for 5.5 years on average; approximately 38% were in rural areas. About half of students were female, 27% had enrolled within six months of the intervention, and 65% had been enrolled for at least one year. In Liberia, the baseline is a reading pre-test administered before the intervention.

Endline literacy assessments were administered after the intervention in both experiments. In Kenya, students in different grades took grade-specific tests; we standardize all scores to mean zero and unit standard deviation in the control group within each grade. Appendix Tables A1 and A2 document the sample construction. Table 1 reports summary statistics for Liberia and Table 2 for Kenya. Figure 2 plots baseline score distributions by grade and treatment status; the substantial overlap across adjacent grades shows why a cross-grade score rule is administratively relevant, while the signal-quality diagnostics below ask whether that score is predictive enough to improve assignment relative to grade.

3.5 Balance and Attrition

Table 3 reports covariate balance across treatment and control units. Baseline test scores are balanced in both experiments. The Kenya balance table shows smaller pre-treatment class sizes in treatment schools, while the Liberia class size differences are imprecise; we therefore treat class size comparisons as part of the classroom-reallocation evidence rather than as balanced baseline covariates.

Endline scores are observed for 3,154 of 4,784 Liberia analytic-sample students (65.9%) and 4,954 of 7,102 Kenya analytic-sample students (69.8%; Appendix Tables A1 and A2). Tables 4 and 5 report missing-endline regressions. Liberia attrition is nearly identical across arms. In Kenya, raw attrition is 22.9% in treatment schools and 31.4% in control schools; with strata fixed effects, treatment students are 7.5 percentage points less likely to be missing an endline score (SE = 4.5 percentage points). Attrition is therefore not perfectly balanced; the main specifications condition on observed endline outcomes and adjust for baseline achievement and strata.

4 Empirical Strategy

4.1 Intent-to-Treat Effects

We estimate the overall treatment effect using:

$$Y_i = \alpha + \beta T_j + \gamma \hat{\theta}_i^{EB} + \phi_s + \varepsilon_i \quad (14)$$

where Y_i is the standardized endline score, T_j is treatment assignment for the relevant randomization unit, S_j denotes the randomization stratum, ϕ_s are stratum fixed effects, and standard errors are clustered at the assignment-unit level: school in Kenya and school grade group level in Liberia. Because treatment was randomized within strata, β is the observed-endline-sample intent-to-treat estimand; interpreting it as the full analytic-sample effect additionally requires that outcome observation is not selectively related to potential outcomes after conditioning on the controls. We therefore report attrition diagnostics and Lee bounds below. We also estimate effects by grade group and by track assignment (upper vs. lower), interacting treatment with the relevant indicators.

The baseline control $\hat{\theta}_i^{EB}$ is an empirical Bayes ability measure that shrinks each student’s raw score toward the grade mean by the grade-specific signal quality:

$$\hat{\theta}_i^{EB} = \bar{s}_{g(i)} + \hat{R}_{g(i)}^2 (s_i - \bar{s}_{g(i)}), \quad (15)$$

where $\hat{R}_g^2$ is the within grade baseline-endline R^2 in the control group (Section 2). These weights are estimated only among control students, so treatment-group endline outcomes do not enter either the shrinkage weights or the baseline covariate; the treatment coefficient is identified by random assignment, with the EB term serving as a pre-treatment precision control and as the basis for diagnostic classroom measures. For the ITT estimates, the transformation is empirically immaterial: replacing $\hat{\theta}_i^{EB}$ with the raw baseline score leaves the treatment coefficient unchanged to three decimals (Table 22). Because $R_g^2 < \rho_g$ in the model with endline noise, this implementation shrinks more aggressively than the model’s latent posterior, making the resulting ability estimates

conservative.³ Beyond its role as a baseline control, $\hat{\theta}^{EB}$ is used to construct the mismatch measure $\text{Misfit}_i = (\hat{\theta}_i^{EB} - \hat{I}_{k(i)})^2$, where $\hat{I}_{k(i)}$ is the estimated target ability level of track $k(i)$. Grade-specific shrinkage ensures that misfit downweights score variation in grades where the test is uninformative, so that the measure reflects genuine ability dispersion rather than measurement error.

The empirical analysis separates the randomized reduced-form effect from the mechanism and accounting evidence. The ITT estimates are causal effects of offering cross-grade grouping under the implemented rules. The dispersion, assignment, peer/rank, class-size, and accounting exercises characterize post-treatment classroom margins and ask which explanations are consistent with the data. Except where explicitly stated, they should not be read as causal mediation estimates, because track assignment, realized peers, classroom composition, and class size are all consequences of the intervention.

4.2 Effect on Within-Class Dispersion

Before asking whether tracking raises achievement, we ask a prior question: does it actually compress within class ability heterogeneity? The answer depends on signal quality. Tracking sorts students on a noisy signal; if the diagnostic is weak, reassignment is close to random and within class baseline score dispersion falls little.

We measure dispersion using baseline scores rather than endline scores. Baseline scores are pre-determined, so within class baseline dispersion is a purely mechanical function of who the sorting rule places together—it is not affected by any treatment-induced changes in achievement. Using endline scores would conflate the sorting effect with treatment effects on the distribution of outcomes. We estimate:

$$|s_i - \bar{s}_{c(i)}| = \alpha + \beta T_j + \phi_s + \varepsilon_i \quad (16)$$

where $|s_i - \bar{s}_{c(i)}|$ is student i 's absolute deviation from the mean baseline score in her reading classroom $c(i)$. A negative β indicates that tracking compresses within class heterogeneity. Because

³When $\hat{R}_g^2$ is constant across grades, $\hat{\theta}^{EB}$ is an affine transformation of s_i and the mismatch measure is a rescaled version of raw-score misfit. Grade-specific shrinkage matters here because R_g^2 ranges from ≈ 0.03 (Liberia Grade 3) to ≈ 0.56 (Kenya Grade 2). In specifications that also include grade fixed effects, using $\hat{\theta}_i^{EB}$ as a linear baseline covariate is equivalent to interacting the raw baseline score with grade under slopes restricted in proportion to the estimated predictive-power weights. In the main ITT specification, which follows the generated tables and controls for strata rather than separate grade indicators, the EB control also carries the grade-mean component of predicted achievement.

the outcome is itself constructed from the baseline score, this baseline-dispersion specification omits the individual baseline control; the predicted-outcome and endline-dispersion analogs in Table 15 add the baseline control. Figure 3 plots the classroom-level standard deviation of baseline scores in treatment and control classrooms directly.

4.3 Peer/Rank Diagnostic Strategy

When an assignment unit is randomized into tracking, everything about the classroom changes at once: peer composition, instructional match, class size, grade mixing. A naive regression of outcomes on peer mean ability conflates the direct effect of classmates with all of these correlated changes. Studying peer composition separately therefore requires variation that is not driven by other classroom characteristics.

The design offers a source of such variation, but it is not automatically identifying. Consider two students in different treated assignment units who have the same baseline score and enrolled grade and are therefore assigned to the same track under the deterministic rule. Their assigned instructional targets are the same by design, but their classmates differ because the roster composition of each assignment unit differs. One student may end up in a reading group where the other students in her track happen to be relatively strong; another may end up with weaker peers. These roster differences are not chosen by the student or local implementers—they are mechanical consequences of which other eligible students are in the same assignment unit, combined with the exogenous treatment assignment that determines whether and how students are regrouped. They support a peer/rank diagnostic only under the exclusion restriction below: after conditioning on own score, grade, treatment status, and stratum, the score-grade roster of other students in the assignment unit carries no additional information about own unobserved determinants of achievement.

A remaining complication is that students’ *exposure* to this peer composition is nonrandom. Let $P_i = (N_{c(i)} - 1)^{-1} \sum_{\ell \in c(i), \ell \neq i} \hat{\theta}_\ell^{EB}$ denote the leave-self-out mean EB ability of student i ’s classmates in her observed reading classroom $c(i)$. P_i is what Borusyak and Hull (2023) call a *formula* regressor: a known, deterministic function of the treatment assignment T_j and the vector of predetermined baseline ability proxies in the assignment unit.⁴ Although the treatment assignment is random, a

⁴The connection to Borusyak and Hull (2023) is as follows. In their framework, a formula regressor is a known function of exogenous shocks and pre-determined exposure shares, and identification is restored by purging its expected value under the known shock-assignment process. Here, leave-out peer mean P_i is likewise a formula object, but it

student’s realized peer mean depends on her own score—higher-scoring students in treated units are sorted into the upper track and thereby exposed to different peers—so P_i is correlated with own ability and hence with unobserved determinants of achievement.

For the exact design-based check, let $P_i(1)$ and $P_i(0)$ be the peer means student i would face under the treated cutoff rule and under the grade-based control classroom. If p_j is the design probability that assignment unit j is treated, the Borusyak–Hull expectation is

$$\mu_i^{BH} = \mathbb{E}_T[P_i(T_j) \mid \{s_\ell, g(\ell)\}_{\ell \in j}, S_j] = p_j P_i(1) + (1 - p_j) P_i(0). \quad (17)$$

The recentered regressor $\tilde{P}_i = P_i^{obs} - \mu_i^{BH}$ subtracts the component of observed peer exposure mechanically predicted by the known treatment lottery and the assignment unit’s roster. This addresses predictable exposure, but a causal peer/rank interpretation still requires the conditional independence restriction stated below. The main specification below uses a coarser control function approximation, conditioning on treatment interacted baseline score cells rather than constructing μ_i^{BH} for every student. Table 17 then reports the exact Kenya checks using the design-based μ_i^{BH} together with flexible controls in own score interacted with treatment and grade.

Three assumptions underpin this strategy:

Assumption 1 (Random Assignment). $T_j \perp (\theta_i, u_i, \varepsilon_i) \mid S_j$ for all $i \in j$.

Assumption 2 (Conditional Independence). $\{s_\ell, g(\ell)\}_{\ell \in j, \ell \neq i} \perp (\theta_i, \varepsilon_i) \mid s_i, g(i), T_j, S_j$.

Assumption 3 (Scripted Instruction). *The assigned instructional level I_{kg} in each track k is a deterministic function of the track designation and is not deliberately retargeted in response to realized classroom composition.*

depends on both the randomized treatment status T_j and the realized composition of the assignment unit. Because treatment is randomized and track assignment follows deterministic cutoffs, the Borusyak and Hull recentering step is feasible in principle: for each student, one can compute the design-based expected peer mean by averaging over the two classroom states induced by the treatment lottery, conditional on the assignment unit’s observed baseline ability proxy distribution. What their recentering handles is the bias from nonrandom *exposure* to the treatment shock—the fact that a student’s own score determines which track she enters and therefore which peers she faces. What it does not provide is the exclusion restriction needed to interpret the residualized coefficient as a causal peer/rank effect: that requires our Assumption 2, which states that cross-unit variation in classmate composition is uninformative about own unobservables conditional on own score, grade, treatment status, and randomization stratum. This is a social-interactions identifying assumption, not something that follows from the Borusyak and Hull shock-exogeneity condition. We implement the main peer/rank specification approximately, using score-bin fixed effects as a control function, and then report exact design-based control function and recentered checks for Kenya.

Assumption 1 holds by the within-stratum randomization. Assumption 2 is a peer-composition exclusion restriction: conditional on a student’s own score, grade, treatment status, and randomization stratum, the score-grade roster of other students in the assignment unit is uninformative about her unobserved determinants of achievement. This is related to, but distinct from, the Borusyak and Hull (2023) shock-exogeneity condition: they consider settings where an exogenous shock (here, treatment assignment within strata) generates a formula regressor with nonrandom exposure; our Assumption 2 additionally requires that the classmate score-grade roster carries no information about own unobservables beyond what own score, grade, and stratum already capture. Under these two assumptions, the remaining peer variation after an adequate control function—or after exact design-based recentering plus flexible own-score controls—is orthogonal to the outcome error.

We first implement this using a control function approach in the spirit of Borusyak and Hull (2023). We include baseline-decile $\times$ treatment $\times$ grade fixed effects that absorb coarse differences in expected peer composition within observed treatment states, together with stratum fixed effects:

$$Y_i = \zeta P_i + \sum_{d,t,g} \delta_{dtg} \cdot \mathbb{1}\{D_i = d, T_j = t, g(i) = g\} + \phi_s + \varepsilon_i. \quad (18)$$

The decile $\times$ treatment $\times$ grade cells absorb the expected peer composition for each student type up to within-cell variation in s_i , so ζ is estimated from deviations of actual peer composition around that cell-level expectation. It has a causal peer/rank interpretation only if Assumption 2 holds and the discretized control function adequately captures the expected-exposure component. Grade is included in the conditioning because the baseline score distribution differs across grades, so two students in the same decile but different grades face different expected peer compositions. This discretized control function is approximate. We therefore also report exact control function and exact recentered specifications for Kenya, where the deterministic assignment rule and randomization probabilities allow us to construct μ_i^{BH} directly.

Assumption 3 plays a distinct role in distinguishing peer composition from instructional targeting. In most tracking settings, instructional match and peer composition move together: a student placed in a higher track gets both better-matched assigned instruction and different peers. With scripted curricula, assigned instructional level is fixed by track designation. The implemented mis-

match proxy $\hat{M}_i = (\hat{\theta}_i^{EB} - \hat{I}_{k(i)})^2$ is therefore approximately a deterministic function of $(s_i, g(i), T_j)$: it is absorbed by decile $\times$ treatment $\times$ grade fixed effects up to within-decile score variation. This reduces the collinearity between peer composition and assigned instructional match, so that ζ is intended as a peer/rank diagnostic net of mismatch to the extent that the discretization is fine enough and the exclusion restriction is credible.

4.4 Accounting Decomposition

The model decomposes the ATE into four terms (equation 12): assignment gain, peer composition, relative rank, and productive time. The mismatch penalty λ is not separately identified: misfit is a quadratic function of ability, so it is collinear with flexible ability controls in any regression that includes them. The productive-time function $\tau(N_c, V_c^g)$ cannot be estimated from the treatment-induced variation in class size and grade mixing without strong functional-form assumptions. In Kenya, however, administrative records of lesson completion provide an observable proxy for one component of productive instructional time (Section 5). Rather than imposing functional-form assumptions on all channels, we report a descriptive accounting exercise built around the observable classroom shifts and the peer/rank estimates, while treating the latter as specification-sensitive.

The approximate peer/rank regression in Section 4.3 estimates $\hat{\zeta}$, the coefficient on a one-SD increase in leave-out peer mean in the control-function specification. Because peer mean and within class rank move together in tracked classrooms—when classmates get stronger, own rank falls— $\hat{\zeta}$ is best interpreted as a composite peer/rank coefficient rather than a pure peer-level parameter.⁵

The accounting step is separate from the recentering logic. We use $\hat{\zeta}$ to ask how much of the overall ITT would be accounted for by treatment-induced changes in peer composition under a given peer/rank specification. The quantity $\hat{\zeta} \cdot \mathbb{E}[\Delta\bar{\theta}_i]$ —where $\mathbb{E}[\Delta\bar{\theta}_i]$ is the observed mean treatment control difference in leave-out peer means—is the implied contribution of the peer/rank channel under that specification. Comparing this quantity to the overall ITT shows whether the approximate peer/rank channel is large enough to matter descriptively, and how much remains for

⁵The model decomposes the social channel into $\tau_c[\pi \mathbb{E}[\Delta\bar{\theta}_i] + \eta \mathbb{E}[\Delta r_i]]$, where τ_c is the productive-time scaling. Under the approximate peer/rank specification, $\hat{\zeta}$ can be read as $\hat{\zeta} \approx \bar{\tau}_c(\pi + \eta \cdot \partial r / \partial \theta)$, absorbing the average productive-time scaling into the reduced-form coefficient. So $\hat{\zeta} \cdot \mathbb{E}[\Delta\bar{\theta}_i] \approx \bar{\tau}_c[\pi \mathbb{E}[\Delta\bar{\theta}_i] + \eta \mathbb{E}[\Delta r_i]]$ when rank changes are approximately linear in peer-mean changes and the identifying assumptions hold. The accounting exercise uses $\hat{\zeta}$ directly without decomposing it into structural parameters.

assignment gains, productive-time effects, and any other channels. This requires only one estimated parameter ($\hat{\zeta}$) and one observed quantity ($\mathbb{E}[\Delta\bar{\theta}_i]$), avoiding the identification difficulties involved in estimating the full model.

Tables 18 and 21 report the accounting decomposition and the key classroom shifts: absolute EB-to-target deviation, peer composition, class size, and within class baseline score dispersion. Table 11 separately reports squared track target misfit, grade mixing, class size, and peer means by arm. Together, these shifts characterize what regrouping does to classrooms without estimating the structural parameters that govern them.

5 Results

The results follow the two hurdles in the model. We first estimate whether the implemented regrouping rules raised average achievement, then ask whether the rules improved assignment relative to grade and what classroom reallocation they induced. This ordering separates the randomized treatment effects from the diagnostic evidence on why the same broad reform can fail for different reasons.

5.1 Treatment Effects on Test Scores

Table 6 presents the main intent-to-treat estimates. In Liberia, the point estimate is negative but imprecise (-0.212 SD, $p = 0.12$). In Kenya, the point estimate is small and statistically indistinguishable from zero (0.031 SD, $p = 0.56$). The 95% upper confidence bound is about 0.135 SD, below the 0.18 SD gain found by Duflo et al. (2011); the 80% power MDE is 0.15 SD (about 2.8 times the clustered SE). Liberia is less precise by the same calculation (MDE = 0.38 SD), but its negative point estimate motivates the heterogeneity analysis below.

Table 7 breaks the result by assigned track. In Liberia, the lower-track effect is close to zero (-0.019 SD), while the upper-track total is -0.348 SD ($p = 0.03$). The upper-track interaction is negative but imprecise (-0.328 SD, $p = 0.11$), so the table should be read as showing that the negative point estimates are concentrated in the upper track rather than as providing a precise estimate of the interaction itself. In Kenya, the upper-lower split is uninformative: both track-specific effects are close to zero.

5.2 Signal Quality, Assignment, and Classroom Reallocation

Figure 1 summarizes the contrast between the two experiments along the four margins that organize the analysis: baseline predictive content, cutoff compliance, absolute EB-to-target mismatch reduction, and realized class size reallocation.

Table 8 and Figure 6 report baseline predictive power. In Kenya, R_g^2 is 0.509 in Grade 1 and 0.559 in Grade 2. In Liberia, the corresponding values are only 0.034–0.065 across grades. Figure 6 makes the consequence visible: in Kenya, the empirical Bayes posterior remains responsive to the raw score, so the diagnostic carries real information beyond grade; in Liberia, the posterior is pulled so strongly toward the grade mean that the score carries little predictive information beyond enrolled grade. Figure 4 shows the same contrast in raw baseline-endline comovement. This R^2 -based measure blends test precision, achievement persistence, and construct alignment, but the Kenya-Liberia contrast is robust to alternative measures: Table 9 shows that within grade rank persistence from baseline to endline is much higher in Kenya (0.684) than in Liberia (0.242), and that the incremental predictive power of baseline scores over grade fixed effects is an order of magnitude larger in Kenya (0.523 versus 0.050).

One interpretation is that signal quality is not a property of the test instrument alone. It also reflects the institutional environment in which the test is administered and in which prior learning occurred. In Kenya, NewGlobe operated its own private-school network, building and running schools under a highly centralized model with control over curriculum, teacher training, and school operations. In Liberia, by contrast, NewGlobe managed existing public schools under a public-private reform: schools remained government-owned, free to students, non-selective, and staffed by civil-service teachers on the government payroll. The Kenya-Liberia contrast is consistent with the idea that less standardized instructional histories and weaker classroom systems can make a one-time diagnostic less predictive of later achievement. We do not isolate these institutional channels experimentally, but the comparison cautions against treating diagnostic accuracy as independent of the production environment in which the signal is generated.

Table 10 and Figure 7 confirm that treated students’ observed track assignments in both experiments follow the published cutoffs exactly, with zero observed assignment error. In Liberia, observed assignments follow the diagnostic rule precisely in the analytic sample, but the rule is

based on a signal that carries little information.

Table 11 and Figure 8 show what reassignment does to classrooms. Regrouping lowers squared track target misfit in both settings: mean squared misfit falls from 1.35 to 0.42 in Kenya and from 1.52 to 0.64 in Liberia. The classroom consequences, however, diverge. In Kenya, treatment reading classes are smaller (18.0 versus 20.9 students), but baseline imbalance (Table 3) makes this contrast descriptive; grade mixing is the clearer treatment-induced change. In Liberia, treated reading classes grow substantially (32.2 to 42.2 students) and also become more mixed by grade. Figure 3 shows that the compression in within class baseline score dispersion is also much stronger in Kenya than in Liberia. The regression analog in Table 15 gives the same pattern: treatment reduces each student’s absolute deviation from the classroom mean by 0.257 standardized-score units in Kenya but only 0.022 in Liberia, and the contrast is similar for predicted outcomes.

The Kenya setting therefore lets us ask a sharper question than whether the score was predictive: did the assignment margin created by the rule have revealed payoff? Table 12 reports three checks. First, students whose reading group changes under treatment do not benefit more than stayers: the mover ITT is 0.030 SD, the stayer ITT is 0.035 SD, and the treatment-by-mover interaction is essentially zero. Second, among movers, being farther from the relevant cutoff—and therefore more clearly assigned to the new track under the rule—does not predict larger treatment gains. The treatment-by-distance slope is 0.005 for Grade 1 students moved up and -0.006 for Grade 2 students moved down. Third, an omnibus predicted-assignment-gain measure, defined as the reduction in squared EB-to-target mismatch from replacing grade assignment with treatment-track assignment, also does not forecast treatment-effect gains: the interaction with treatment is 0.008 SD per one standard deviation of predicted assignment gain ($SE = 0.068$). A complementary descriptive check in control classrooms reaches the same conclusion: students farther from their grade-level target do not have meaningfully lower endline scores once school-grade mean baseline achievement is controlled for. These tests do not show that correct placement is irrelevant in all settings. They show that the particular cross-grade margin created by the Kenya cutoff rule has little revealed payoff in these data, even though the rule improves measured sorting.

5.3 Treatment Effect Heterogeneity

The negative Liberia upper-track estimate could reflect latent misassignment: students just above the cutoff may truly belong in the lower track, and placing them in the upper track may lower their achievement. Table 13 tests this by breaking upper-track students into within-track ability terciles. If this channel were central, negative estimates should be concentrated in the bottom tercile—students most likely to truly belong in the lower track. Instead, the point estimates are largest for the middle (-0.472 SD, $p = 0.06$) and top (-0.354 SD, $p = 0.02$) terciles, while the bottom tercile effect is smaller (-0.204 SD, $p = 0.24$). Table 14 points in the same direction from another angle: the largest negative estimates appear in the two quintiles farthest from the assignment cutoff, rather than among students closest to the cutoff (Figure 5). The continuous distance interaction is negative but imprecise, so this evidence is suggestive rather than a sharp test of monotonicity.

This pattern is more consistent with constraints in the upper-track classroom than with negative estimates concentrated among marginally misassigned students. Table 11 shows that regrouping increases mean reading class size in Liberia from 32.2 to 42.2 students while increasing mixed grade exposure. Adding class size as a control attenuates the upper-track interaction from -0.328 ($p = 0.11$) to -0.237 SD ($p = 0.19$), but the total upper-track effect is almost unchanged (Table 25). Since class size is itself an outcome of regrouping and is tightly linked to track assignment, this conditional specification is best read as a post-treatment residualization check: it shows that the upper-track pattern is not mechanically eliminated by adding realized class size, but it does not identify the separate causal contribution of class size. The number of comparisons grows as we cut by track, grade block, and distance from cutoff, and the cell sizes shrink accordingly, so these patterns should be read as suggestive.

5.4 Peer/Rank Diagnostics

In Kenya, score-based sorting compresses within class baseline score dispersion (Figure 3) but does not generate evidence of average score gains. A natural way to probe the classroom-reorganization interpretation is to ask whether regrouping changes peer composition or relative rank in directions large enough to offset the measured sorting improvement. We therefore use the approximate

peer/rank specification in Section 4.3 as a diagnostic: first asking whether the approximate control function regression delivers an association large enough to matter, and then asking whether that estimate survives exact control function and recentering checks.

Table 16 reports the approximate decile-cell diagnostic estimates. In Liberia, the peer-mean coefficient is small and imprecise ($\hat{\zeta} = -0.048$, $p = 0.66$). In Kenya, the approximate specification estimates that a one-SD increase in peer mean EB ability is associated with 0.133 SD lower own achievement ($p = 0.02$). Because the regressor is only approximately purged by score cells, this estimate is a first-pass channel diagnostic rather than a final causal peer/rank estimate. It is stable when deciles are replaced by ventiles, but it is not robust to the exact checks in Table 17. When the design-based expected peer regressor μ_i^{BH} is used either as a control function or to recenter realized exposure, the coefficient on realized or recentered peer exposure is close to zero: 0.001 in the exact control-function specification and -0.009 in the exact recentered specification. The nonzero coefficient on μ_i^{BH} in the exact control-function columns is a coefficient on predictable exposure, not on the residual peer/rank variation used for identification; it is therefore not a causal peer/rank coefficient. It captures the component of peer exposure mechanically induced by the assignment rule and roster composition.

The peer evidence should therefore be interpreted cautiously. The approximate specifications show that adverse peer/rank movements are large enough to matter relative to the small Kenya ITT in accounting terms, but the exact specifications do not support interpreting those movements as a robust causal peer/rank effect. The implication is not that classroom reorganization is irrelevant; it is that the Kenya null should not be attributed to one identified peer/rank coefficient. The robust fact is descriptive and central to the paper: Kenya’s assignment rule changed the classroom margins it was designed to change—baseline dispersion, peer exposure, and rank positions—without producing average gains.

Probing peer-level versus rank effects. Conditional on the approximate peer/rank specification, the composite $\hat{\zeta}$ captures both a peer-level effect (π) and a rank effect (η). Under the model, $\hat{\zeta}(\theta_i) \approx \pi + \eta \cdot (-f(\theta_i))$, where $f(\theta_i)$ is the density of classmates’ baseline scores near student i ’s own score. Students in the tails of the ability distribution have few nearby classmates, so their within class rank is insensitive to changes in peer composition; students in the middle have many nearby

classmates, so rank responds more strongly to peer shifts. If rank effects matter ($\eta \neq 0$), then $\hat{\zeta}$ should be more negative in high-density regions. Table 20 tests this by estimating the approximate peer/rank regression separately for terciles of baseline score density and by interacting peer mean EB ability with standardized density. The tercile estimates are negative in all three bins (-0.179 , -0.099 , -0.123), and the pooled peer-by-density interaction is small relative to its standard error (-0.024 , $SE = 0.039$). Within the approximate specification, we do not find that $\hat{\zeta}$ becomes more negative in the high-density middle of the score distribution. Because the exact peer/rank checks are close to zero, this exercise should be read as a diagnostic on the approximate estimate rather than as evidence that rank effects are absent.

5.5 Accounting Decomposition

We now apply the descriptive accounting exercise described in Section 4.4, using the approximate $\hat{\zeta}$ and the observed peer shifts to gauge the scale of the peer/rank component relative to the overall ITT. This is not a causal decomposition: $\hat{\zeta}$ is a reduced-form composite of peer-level and rank effects, the mapping from peer-mean shifts to rank shifts is an approximation, and the exact-BH specifications substantially attenuate the peer/rank coefficient. The exercise asks whether the approximate peer/rank component is large enough to be relevant, not whether it causally explains the small Kenya ITT.

Table 18 reports the Kenya results. In the approximate specification, the peer/rank estimate is $\hat{\zeta} = -0.133$ ($p = 0.02$), and treatment raises mean leave-out peer EB ability by about 0.28 SD. The implied peer/rank contribution is about -0.04 SD. This magnitude is large enough to matter relative to the small Kenya ITT. However, because the exact-BH estimates are close to zero, the calculation should be treated as an upper-bound diagnostic on one possible mechanism rather than the preferred causal decomposition.

The descriptive classroom shifts in the same table show that treatment reduces the absolute EB-to-target deviation by 0.421 SD and compresses within class baseline score dispersion (0.326 SD), so the sorting rule changes the measured classroom margins it was designed to change.

Table 19 provides a check on one observable component of the productive-time channel. In the model, productive time τ_c is a multiplicative scaling factor on all instructional channels (equation 6), so the productive-time contribution to the treatment effect is $[\tau_{c(i,T)} - \tau_{c(i,G)}] \cdot A_i(T)$, where

$A_i(T)$ collects the within-bracket instructional terms under treatment. The administrative measure is narrower than this model object: it records the fraction of scripted literacy-revision lessons completed, not total effective instructional time. The point estimates are positive: treatment raises the school–subject–grade lesson-completion rate by 1.8–2.4 percentage points across specifications. With school clustered standard errors, however, these estimates are not statistically distinguishable from zero. The evidence therefore does not support a large negative productive-time shock in Kenya, but it also cannot establish a precise positive productive-time gain. Even taking the largest point estimate literally, the control-group completion rate is approximately 17%, so a 2.4 percentage-point increase would imply a roughly 15% proportional increase in this measured completion margin. Under the illustrative assumption that a full school year of learning in comparable settings is on the order of 0.2–0.4 SD (e.g., Muralidharan et al., 2019) and one term generates roughly a third of that, the implied contribution would be only 0.010–0.020 SD if lesson completion scaled overall productive time one-for-one.⁶ Thus the lesson-completion evidence does not resolve why Kenya’s measured sorting changes do not translate into average score gains.

Table 21 reports the Liberia results. The approximate decile-cell peer/rank coefficient is near zero ($\hat{\zeta} = -0.048$, $p = 0.66$), and the treatment-induced peer shift is small (0.041 SD), so the peer/rank contribution is negligible. The near-zero peer/rank coefficient and small classroom composition shift provide little evidence that the measured peer/rank channel explains the negative Liberia ITT pattern. Taken together with the negative overall ITT point estimate, the larger reading classes, and greater grade mixing, the evidence is more consistent with broader reorganization constraints than with the measured peer/rank channel, but it does not isolate which reorganization margin is causal.

The Kenya-Liberia comparison illustrates two different constraints on cross-grade grouping, but it does not isolate signal quality from other institutional differences—the two settings differ in school governance, teacher selection, class size constraints, and student composition, all of which could independently affect the returns to regrouping.

⁶Because the administrative data record completion of supplementary revision lessons rather than the full scripted curriculum, this calculation should be read only as a scale check on the measured completion margin. If the measure captures only this narrow slice of instruction rather than overall classroom productivity, the implied contribution is an order of magnitude smaller (0.002–0.003 SD).

6 Robustness

6.1 Specification and Inference

Table 22 reports the ITT estimates under alternative specifications—raw baseline score instead of EB ability as a control, models without any baseline control, wild cluster bootstrap p values, and permutation-based inference at the country-specific assignment unit. Replacing EB ability with the raw baseline score leaves the estimates unchanged to three decimals. Dropping the baseline control moves the Kenya point estimate upward (0.110, SE = 0.071) and leaves Liberia negative (−0.240, SE = 0.139). Baseline adjustment therefore affects both precision and, especially in Kenya, the point estimate. Even so, the robustness checks do not overturn the main interpretation: the adjusted Kenya estimate remains small and imprecise, while the Liberia estimates remain negative. The bootstrap and permutation p values are close to the clustered-inference results.

Because Kenya attrition is modestly differential, Table 23 reports Lee point-estimate bounds for the main ITT. Kenya treatment schools have lower attrition than control schools, so the Lee procedure trims the treatment outcome distribution to equalize response rates. The resulting Kenya point-estimate bounds are $[-0.023, 0.154]$ SD. Thus selective attrition could move the Kenya estimate modestly in either direction, but even the upper point-estimate bound remains below the 0.18 SD benchmark from within grade tracking. Liberia attrition is essentially balanced across arms, so the Lee bounds coincide with the OLS estimate.

6.2 Data-Driven Treatment-Effect Heterogeneity

The subgroup results above are guided by the model’s structure: upper versus lower track, distance from the cutoff, peer-composition channels. A natural concern is that these are researcher-chosen, low-dimensional cuts and may not capture the most policy-relevant dimensions of heterogeneity. We therefore apply the generic machine-learning inference framework of Chernozhukov et al. (2025) as a data-driven stress test: does a flexible learner, unconstrained by the model’s partition, find stable evidence that treatment effects vary in policy-relevant ways beyond the average?

We estimate an honest causal forest (Wager and Athey, 2018) separately for each experiment, using only pre-treatment covariates as predictors: baseline test scores, empirical Bayes ability

estimates, grade, the deterministic upper-track indicator implied by the baseline rule, distance from the track cutoff, and—for Kenya—historical achievement records from prior terms. Treatment is randomized at the assignment-unit level; all inference respects this clustering. Predicted treatment effects for the sorted groups are based on cluster-level cross-fitting to avoid overfitting. We sort students into quintiles of predicted effects and estimate reduced-form effects within each quintile (sorted GATES), then test for overall heterogeneity using the best linear projection (BLP).

Table 24 and Appendix Figure A5 report the results. Groups are sorted by the forest-predicted treatment effect, with Group 1 containing students with the lowest predicted effects and Group 5 those with the highest. In Kenya, the realized estimates reverse this ranking: Group 5 has a negative estimated effect (-0.180 SD, $p < 0.05$), while Group 1 has a positive but imprecise estimate (0.238 SD). The Group 5-minus-Group 1 contrast is therefore negative (-0.418 SD, $p < 0.10$), but the BLP test does not reject constant effects at conventional levels ($p = 0.13$). The forest does not provide stable evidence for a targeting rule; the reversed realized ranking illustrates how noisy individual-level treatment-effect prediction is in this setting.

In Liberia, the BLP test also does not reject constant effects at conventional levels ($p = 0.13$). The quintile estimates are all negative except for the second quintile, and the most negative point estimates occur in the first and fifth quintiles. Although both endpoint estimates are negative, their contrast is imprecise, so the forest does not reveal a simple monotone targeting rule. This pattern is consistent with the broader evidence that Liberia’s low-signal diagnostic offers little support for fine-grained assignment decisions, while also reflecting the limited precision of the heterogeneity exercise.

Realized reading class sizes differ across experiments: treatment classes are smaller in Kenya (18.0 versus 20.9 students), a descriptive contrast given baseline imbalance (Table 3), and substantially larger in Liberia (42.2 versus 32.2 students). The Liberia increase more directly reflects the mechanics of combining two grades’ worth of students into two tracks within existing school infrastructure. This raises the concern that the Liberia results are mainly an overcrowding artifact rather than a feature of the broader classroom reorganization the model emphasizes. Table 25 reports the track-interaction specification with and without class size controls. The total upper-track effect in Liberia is essentially unchanged after adjustment (-0.348 without the control and -0.345 with it), and the class size coefficient itself is small. In Kenya, adding class size leaves the track coefficients

essentially unchanged. Because realized class size is post-treatment, these estimates should not be interpreted as a causal mediation exercise or as evidence that class size is irrelevant; they show only that the main treatment-effect patterns are not mechanically eliminated by conditioning on realized class size.

6.3 Measurement and Distributional Concerns

The small Kenya ITT is the paper’s central puzzle: score-based sorting that compresses baseline score dispersion but yields no evidence of average score gains. Two features of the measurement system could spuriously attenuate the average effect, so it is important to check whether either is consistent with the data.

First, the composite score has a 50-point maximum, and the upper tail of both baseline and endline distributions reaches that ceiling. If treatment genuinely raises achievement for high-ability students but their gains are censored at the maximum, the measured ITT would be attenuated toward zero. Importantly, this concern cuts *against* our interpretation: ceiling effects would bias toward finding a small average effect even if tracking helps, so they make the small Kenya estimate conservative rather than artifactual. They would also work against finding a negative upper-track effect, which we do not find in Kenya. Table 26 shows that the point estimate remains close to the baseline OLS estimate when estimated by Tobit with an upper limit at 50 points (0.040 SD) or by OLS after dropping the top baseline score decile (0.022 SD); 176 endline observations reach the ceiling.

Second, tracking mechanically reshuffles students across classrooms, which could compress or expand the overall distribution of scores even if the mean is unchanged. The concern is specific: if treatment compresses the distribution—pulling high scores down and low scores up—a near-zero average effect might mask real changes that cancel out across the ability distribution, and the standard ITT would miss a genuine distributional effect. Table 27 provides little evidence of such a total-variance effect in Kenya: the adjusted treatment coefficient on total variance is small and imprecise (0.038 SD², SE = 0.073). At the classroom level, tracking significantly reduces within class variance (−0.228 SD², $p < 0.01$) and increases between class variance by almost exactly the same amount (0.239 SD², $p < 0.01$). The sorting rule therefore redistributes achievement variance across classrooms—precisely what tracking is designed to do—but does not meaningfully compress

or expand the total score distribution. This does not point to distributional reshuffling as the explanation for the small Kenya ITT.

7 Conclusion

Cross-grade ability grouping did not generate evidence of average achievement gains in either setting. The diagnostics point to two distinct constraints. In Liberia, observed track assignments followed the published cutoffs exactly and reduced measured distance to implemented track targets, but the diagnostic carried little endline-predictive information, the resulting classrooms remained heterogeneous on predictive dimensions, and reading class sizes rose by roughly one-third. In Kenya, the diagnostic was highly predictive and grouping substantially reduced within class baseline score dispersion, but these measured sorting changes are not accompanied by evidence of higher average achievement. Nor do the students most directly affected by the assignment rule—movers, students farther from the cutoff among movers, or students with larger predicted assignment gains—show larger treatment effects.

The two experiments therefore suggest a two-part test for assignment rules. First, the diagnostic must contain enough information to improve on the incumbent rule. Second, the organization must be able to act on that information without creating a less productive environment. Grade-level assignment is coarse, but it bundles age, developmental stage, curricular exposure, and peer cohorts. Replacing it with a diagnostic assessment separates these elements: students are mixed across ages, peer groups shift, and class sizes change. In the settings studied here, neither Kenya’s stronger predictive signal nor Liberia’s mechanical movement toward implemented track targets was accompanied by average achievement gains when those same rules also reorganized classrooms along peer, grade-mixing, and class-size margins.

The decomposition clarifies what must be explained. Regrouping simultaneously alters instructional mismatch, peer ability, rank, and productive instructional time. When the diagnostic is weak, the rule may still reduce distance to its own track targets, but those targets are built from a signal that carries little information about later achievement; schools can therefore incur the logistical costs of reorganizing without creating classrooms that are much more homogeneous on predictive dimensions. When the diagnostic is strong, placement on the measured signal still does

not guarantee higher achievement: improving the assignment margin is only one input into learning, and the resulting classroom environment may fail to convert better placement into higher scores. In Kenya, the assignment-payoff tests make this point directly. The rule changes who is assigned where, but treatment gains are no larger for the students who should benefit most from the new placement under the implemented rule.

The findings also caution that diagnostic accuracy should not be treated as a freely chosen policy lever. In environments with less standardized instruction, limited teacher training, and heterogeneous curricular exposure, a single diagnostic may be less predictive because the underlying system generates wide within grade dispersion and less stable links between current scores and later achievement. Cross-grade grouping is often most attractive in these contexts, but it is also harder to execute. It demands that the diagnostic produce comparable rankings across grades and forces administrators to absorb the resulting grade-mixing and class size reallocations. By contrast, within grade tracking avoids these additional requirements. The positive effects documented by Duflo et al. (2011) for within grade tracking in Kenya are consistent with this distinction.

The problem extends beyond schools. In many assignment settings—algorithmic risk scores for judicial discretion, standardized assessments for teacher judgment, credit models for relationship banking—a finer signal replaces a coarse but entrenched rule embedded in a production environment that has adapted to it. The two experiments illustrate why a more informative diagnostic need not guarantee better outcomes once the reorganization it induces is taken seriously. The policy implication is to look for evidence on both thresholds: whether a new signal materially improves classification, and whether it does so without making the reorganized production process less productive than the status quo.

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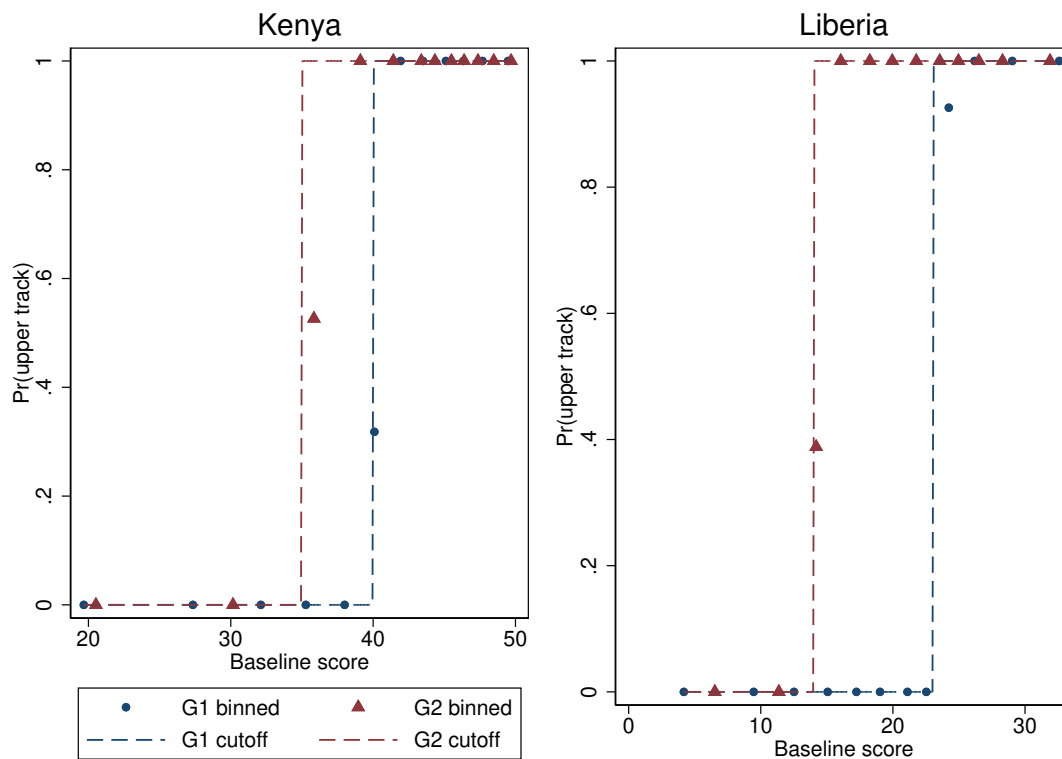
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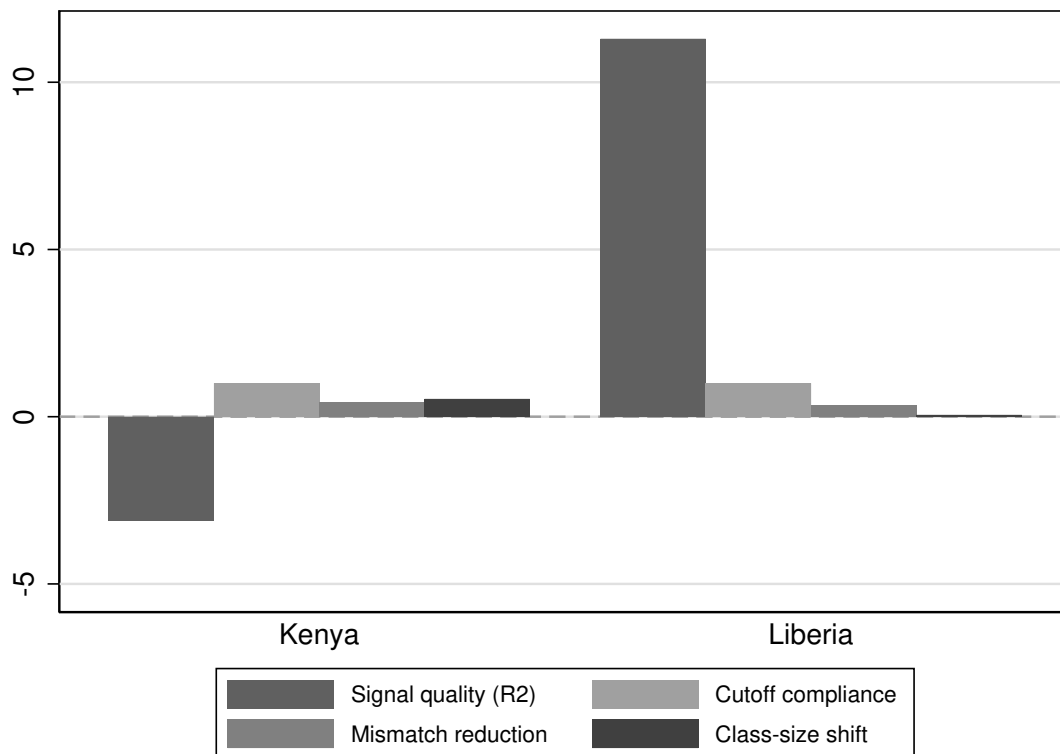
Figures

Figure 7: Deterministic Assignment Cutoffs



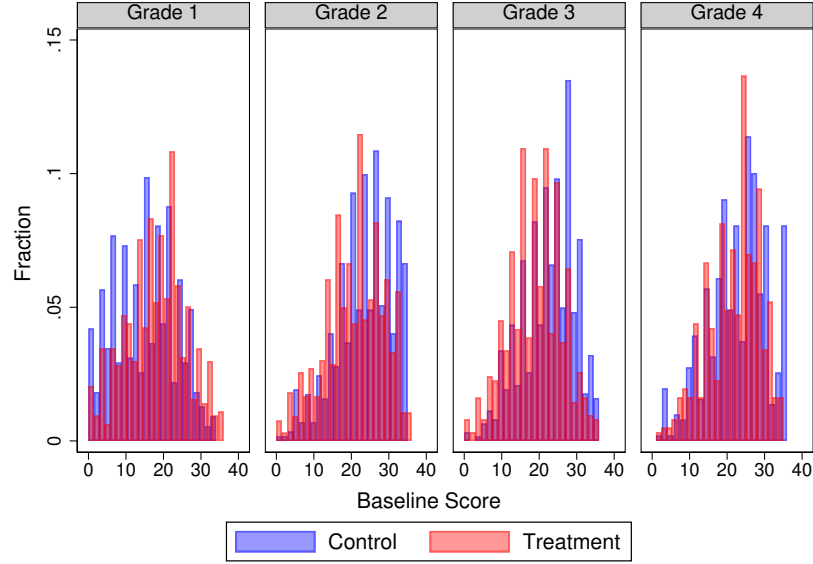
Notes: This figure plots the empirical probability of assignment to the upper reading track against the raw baseline score for students in treated units. Panels correspond to the published assignment rules: grade-specific cutoffs in Kenya and grade group cutoffs in Liberia. The discontinuous jumps occur exactly at the published score thresholds, showing that observed track assignments follow deterministic cutoffs with no observed assignment error in the analytic sample.

Figure 1: Kenya-Liberia Contrast on Four Assignment Margins

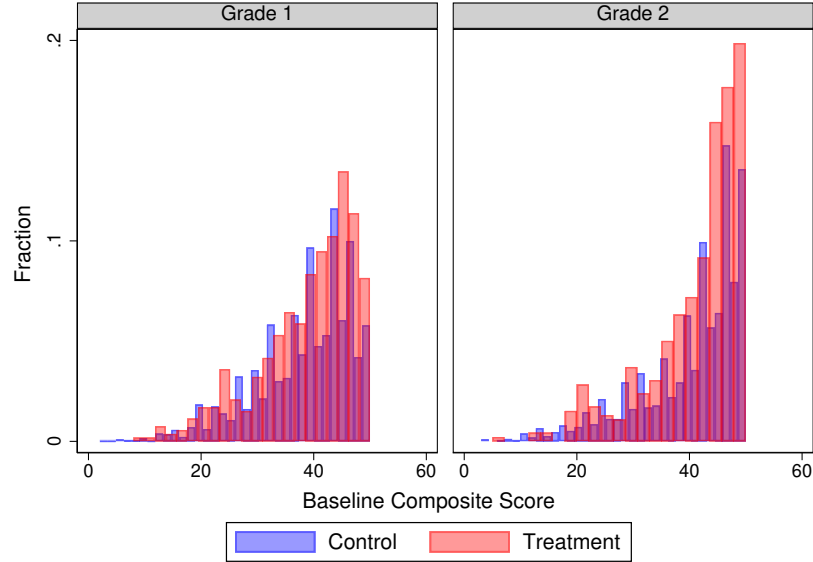


Notes: This summary figure compares Kenya and Liberia along the four margins that organize the paper. Panel (a) reports weighted baseline predictive content (R_g^2), Panel (b) reports cutoff compliance: the share of observed track assignments that exactly follow the published deterministic cutoff rule, Panel (c) reports absolute EB-to-target mismatch reduction (control minus treatment means), and Panel (d) reports realized reading class size differences (treatment minus control means). In both settings, observed treated-student assignments follow the deterministic cutoffs and measured misfit falls, but Kenya has much stronger predictive content and smaller realized treatment classes, while Liberia has weak predictive content and much larger treatment classes.

Figure 2: Baseline Score Distributions



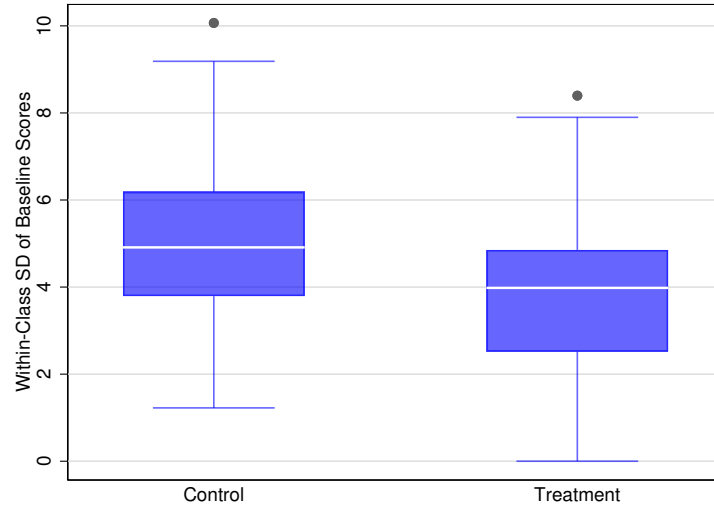
(a) Liberia



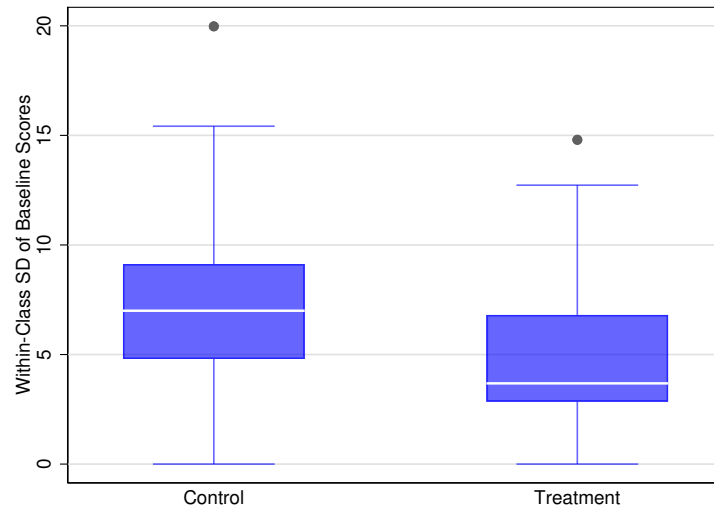
(b) Kenya

Notes: This figure plots kernel density estimates of raw baseline reading scores for students in the analytic sample, separately by study, enrolled grade, and treatment assignment. The vertical dashed lines mark the score cutoffs used to assign students to the upper reading track in treated units. The substantial overlap across adjacent grades shows why a cross-grade score rule is administratively relevant in both settings; the signal-quality diagnostics ask whether that score is predictive enough to improve assignment relative to grade.

Figure 3: Within-Class Dispersion: Treatment vs. Control



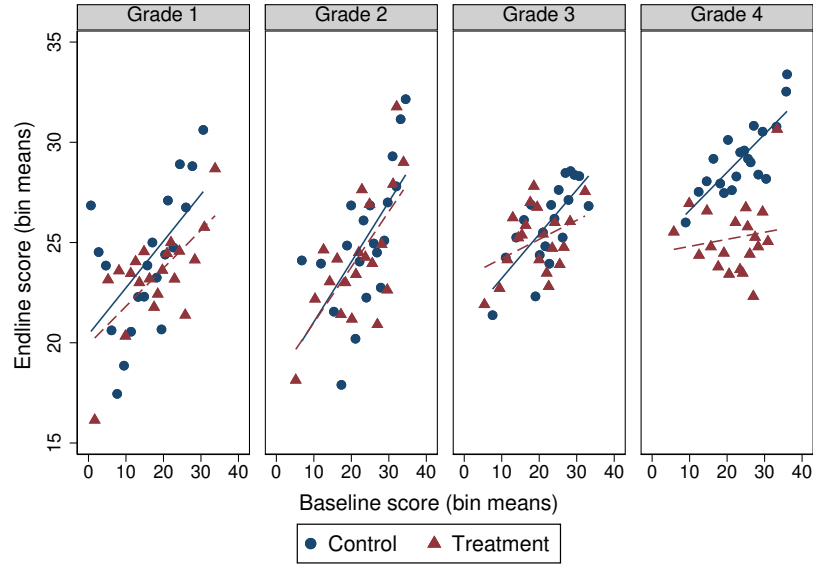
(a) Liberia



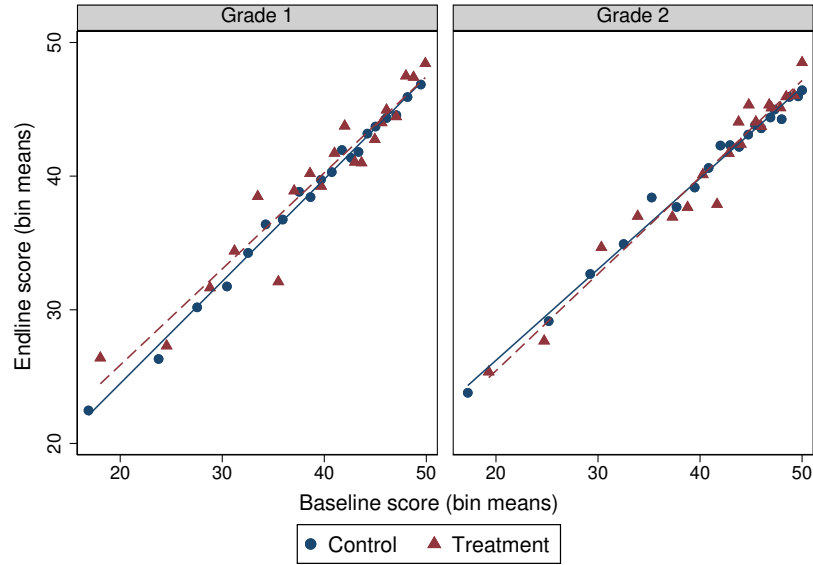
(b) Kenya

Notes: Each panel shows the distribution of the classroom-level standard deviation of baseline reading scores for realized reading classrooms in treatment and control units. The unit of observation is a reading classroom. A leftward shift indicates greater within class homogeneity. The compression is stronger in Kenya than in Liberia.

Figure 4: Baseline vs. Endline Scores



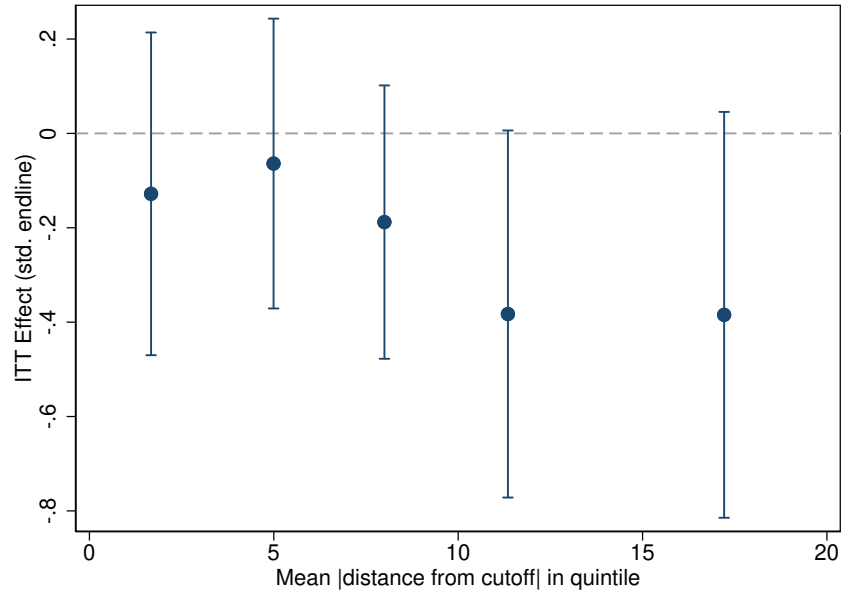
(a) Liberia



(b) Kenya

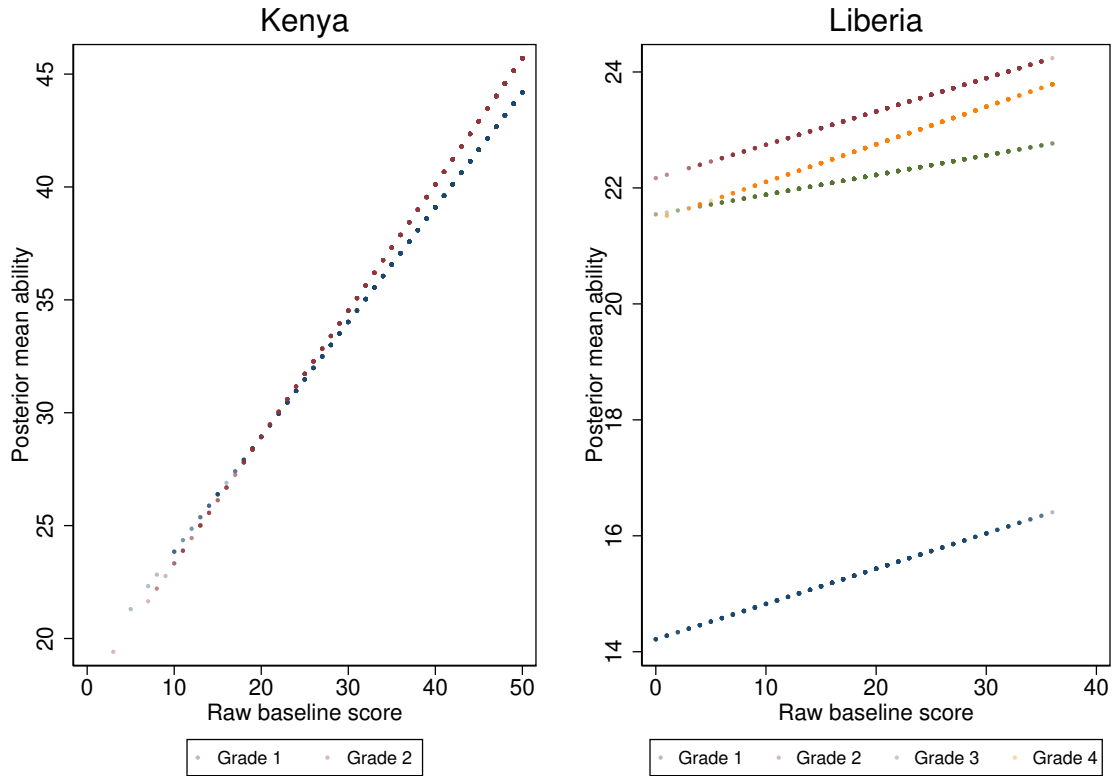
Notes: This figure plots binned means of endline reading scores against baseline reading scores for students in the analytic sample, separately by study, enrolled grade, and treatment assignment. Higher slope indicates that baseline achievement is more informative about later achievement. The stronger baseline-endline comovement in Kenya than in Liberia mirrors the difference in baseline predictive power reported in Table 8.

Figure 5: Treatment Effect by Distance from Assignment Cutoff in Liberia



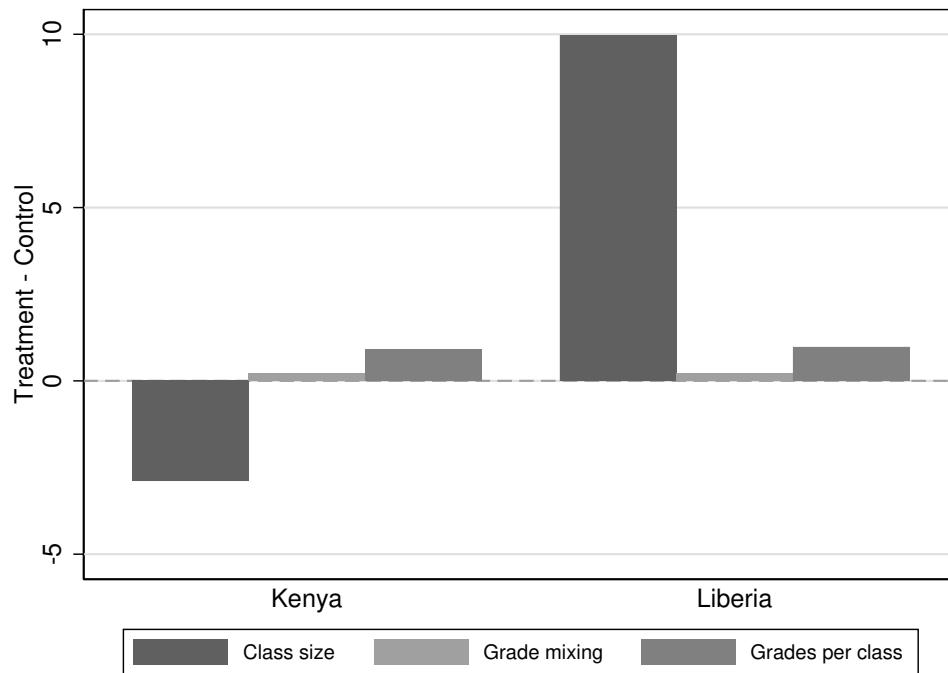
Notes: Bars report Liberia intent-to-treat estimates for standardized endline reading scores, separately by quintiles of a student's absolute distance from the grade group assignment cutoff. The underlying estimates come from student-level OLS regressions with strata fixed effects and the empirical Bayes baseline control used elsewhere in the paper; error bars denote 95% confidence intervals based on standard errors clustered at the school grade group level. If negative estimates came mainly from students being marginally misassigned in latent ability near the threshold, the largest negative estimates would appear in the leftmost bins. Instead, the largest negative estimates appear in the two bins farthest from the cutoff, while the near-cutoff bins are smaller and imprecise.

Figure 6: Posterior Shrinkage and Signal Quality



Notes: The top panels show the distribution of raw baseline reading scores by grade in each study. The bottom panels map the empirical Bayes posterior mean of latent ability to the raw baseline score using the grade-specific baseline predictive power estimates in Table 8. When signal quality is low, the posterior is pulled strongly toward the grade mean; when signal quality is high, the posterior tracks the raw score more closely. Liberia therefore exhibits much stronger shrinkage than Kenya.

Figure 8: Treatment-Induced Classroom Reallocation



Notes: Bars show treatment minus control differences in classroom margins measured from realized reading classrooms in the structural analysis sample. The three outcomes are mean class size, mean within class variance of enrolled grade, and mean number of grades represented in the reading classroom. Positive values indicate that treatment increases the corresponding margin. Treatment-arm reading classes are smaller in Kenya and larger in Liberia, and grade mixing increases in both settings.

Tables

Table 1: Summary Statistics — Liberia

Variable	Experiment (Liberia)			
	(1)	(2)	(3)	(4)
	Control Mean	Treatment Mean	SD (All)	N
Baseline score	21.014	19.848	7.936	4784
Endline score	25.983	24.467	7.612	3154
Upper-track assignment	0.612	0.549	0.494	4784
Has endline	0.659	0.660	0.474	4784
Class size	45.596	59.278	27.288	4784

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table 2: Summary Statistics — Kenya

Variable	Experiment (Kenya)			
	(1)	(2)	(3)	(4)
	Control Mean	Treatment Mean	SD (All)	N
Baseline score	39.207	39.904	8.786	7102
Endline score	39.333	40.258	8.519	4954
Upper-track assignment	0.614	0.652	0.486	7102
Has endline	0.686	0.771	0.459	7102
Class size	24.307	20.935	11.233	7102

Notes: Means and standard deviations are reported in the analytic sample. Endline outcomes are based on observations with non-missing endline test scores.

Table 3: Baseline Balance

	Experiment 1 (Kenya)			Experiment 2 (Liberia)		
	Control#mean (1)	Coef. (2)	N (3)	Control#mean (4)	Coef. (5)	N (6)
Class size	19.094	-3.949*** (1.261)	213	22.357	2.652 (2.497)	99
Lower grade class size	19.697	-3.563** (1.387)	212	23.286	1.377 (2.630)	99
Upper grade class size	18.528	-4.370*** (1.292)	213	21.224	3.158 (2.830)	99
Baseline test score	0.037	0.040 (0.074)	213	-0.061	-0.150 (0.118)	99
Lower grade baseline score	0.010	0.096 (0.098)	212	0.030	-0.127 (0.155)	99
Upper grade baseline score	0.066	-0.018 (0.075)	213	-0.120	-0.198 (0.128)	99

Notes: Kenya specifications are at the school level and control for constituency fixed effects, with standard errors clustered at the school level. Liberia specifications are at the school grade group level and control for randomization strata fixed effects, with standard errors clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 4: Attrition — Liberia

	Experiment (Liberia)			
	(1)	(2)	(3)	(4)
Sample	T missing	C missing	Adj. diff.	N
Overall	0.340	0.341	0.000 (0.062)	4784
Grade 1	0.328	0.249	0.079 (0.093)	1184
Grade 2	0.334	0.301	0.033 (0.115)	1233
Grade 3	0.322	0.479	-0.157 (0.108)	1243
Grade 4	0.379	0.316	0.063 (0.109)	1124

Notes: Attrition = missing endline. T missing and C missing are unadjusted arm means. Adjusted difference is the treatment coefficient from an OLS regression with strata FE; SEs clustered at the school grade group level.

Table 5: Attrition — Kenya

Sample	Experiment (Kenya)			
	(1)	(2)	(3)	(4)
	T missing	C missing	Adj. diff.	N
Overall	0.229	0.314	-0.075 (0.045)	7102
Grade 1	0.233	0.291	-0.054 (0.052)	3618
Grade 2	0.225	0.337	-0.088 (0.058)	3484

Notes: Attrition = missing endline. T missing and C missing are unadjusted arm means. Adjusted difference is the treatment coefficient from an OLS regression with strata FE; SEs clustered at the school level.

Table 6: Effect of Ability Grouping on Test Scores

	Experiments	
	(1)	(2)
	Kenya	Liberia
Treatment	0.031 (0.053)	-0.212 (0.135)
Control Mean	0.012	0.091
N	4954	3154

Notes: Dependent variable is standardized endline score. All specifications control for EB ability and strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 7: Treatment Effects by Track

	Experiments	
	(1)	(2)
	Kenya	Liberia
Lower track (Treatment)	0.046 (0.137)	-0.019 (0.181)
T $\times$ Upper	-0.023 (0.147)	-0.328 (0.201)
Upper track total	0.023 (0.044)	-0.348** (0.154)
N	4954	3154

Notes: Upper track total is Treatment + T $\times$ Upper. Controls: EB ability, strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 8: Baseline-Endline Predictive Power

Study	Control-group diagnostics			
	(1)	(2)	(3)	(4)
	Grade	N (Control)	Corr.	R^2
Kenya	1	2190	0.713	0.509
	2	2005	0.748	0.559
Liberia	1	411	0.247	0.061
	2	399	0.240	0.057
	3	324	0.184	0.034
	4	348	0.255	0.065

Notes: Correlation is the Pearson correlation between baseline and endline raw scores in the control group, by grade. R^2 is the squared correlation and measures the fraction of endline variance predicted by the baseline diagnostic.

Table 9: Alternative Signal-Quality Diagnostics

	Experiments	
	(1)	(2)
	Kenya	Liberia
R-squared grade FE only	0.011	0.059
R-squared grade FE + baseline	0.534	0.109
Incremental R-squared	0.523	0.050
Within-grade rank persistence	0.684	0.242

Notes: All computed in control group. Incremental R-squared = additional variance explained by adding baseline score to a grade FE only model. Rank persistence is the within grade Spearman correlation between baseline and endline score ranks, weighted by grade size.

Table 10: Deterministic Assignment Cutoffs

	Experiments					
	(1)	(2)	(3)	(4)	(5)	(6)
	Kenya			Liberia		
	Cutoff	N treated	Rule error	Cutoff	N treated	Rule error
Grade						
1	40	527	0.000	23	637	0.000
2	35	458	0.000	23	662	0.000
3				14	621	0.000
4				14	615	0.000

Notes: The table reports score thresholds used to assign treated students to the upper reading track. Rule error is the share of treated students whose observed upper-track assignment differs from the published deterministic cutoff rule.

Table 11: Classroom Reallocation under Cross-Grade Grouping

	Experiments			
	(1)	(2)	(3)	(4)
	Kenya		Liberia	
	Control	Treat	Control	Treat
Class size	20.88	17.99	32.19	42.16
Within-class grade variance	0.000	0.209	0.000	0.211
Number of grades in class	1.000	1.896	1.000	1.981
Squared track target misfit	1.352	0.419	1.519	0.639
Peer mean EB ability	-0.006	0.254	0.009	0.032

Notes: Entries are arm-specific means in the structural analysis sample. Realized classroom margins use treated-track classes in treatment and grade classes in control. Squared track target misfit is the square of EB ability minus the track target.

Table 12: Kenya: Assignment Payoff Diagnostics

Measure	Coef.	SE	N
<i>A. Randomized reallocation contrasts</i>			
Movers ITT	0.030	(0.053)	1816
Stayers ITT	0.035	(0.073)	3138
Treatment $\times$ mover	-0.000	(0.082)	4954
<i>B. Assignment intensity among movers</i>			
G1 movers up: Treatment $\times$ cutoff distance	0.005	(0.017)	1286
G2 movers down: Treatment $\times$ cutoff distance	-0.006	(0.023)	530
<i>C. Predicted assignment gain</i>			
Treatment $\times$ predicted assignment gain (1 SD)	0.008	(0.068)	4954
<i>D. Descriptive control-group revealed-value checks</i>			
Mismatch proxy	-0.008	(0.011)	4195
Mismatch proxy + school-grade mean BL	-0.002	(0.011)	4195

Notes: Outcome is the standardized endline test score. Movers are students whose treatment reading-group assignment differs from their enrolled grade: Grade 1 students above the upper-track cutoff and Grade 2 students at or below the lower-track cutoff. Cutoff distance is signed so larger values indicate more clearly assigned movers. Predicted assignment gain is the standardized reduction in squared EB-score mismatch from replacing the grade target with the treatment-track target; the regression also controls for its square. Panel D is estimated only in control classrooms and is descriptive rather than randomized. Standard errors are clustered at the school level.

Table 13: Treatment Effects by Track and Ability Tercile (Liberia)

Track	Ability Bin	Liberia			
		(1)	(2)	(3)	(4)
		ITT	SE	p value	N
Lower	Bottom	-0.192	(0.214)	0.372	460
	Middle	0.150	(0.283)	0.599	459
	Top	-0.055	(0.238)	0.819	393
Upper	Bottom	-0.204	(0.174)	0.244	626
	Middle	-0.472*	(0.242)	0.055	606
	Top	-0.354**	(0.150)	0.021	610

Notes: Within-track ability terciles. OLS with strata FE and EB control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 14: ITT Heterogeneity by Distance from Assignment Cutoff (Liberia)

Quintile	Liberia			
	(1)	(2)	(3)	(4)
	Mean $ s - c $	ITT	SE	N
Q1	1.7	-0.128	(0.174)	813
Q2	5.0	-0.064	(0.157)	636
Q3	8.0	-0.188	(0.148)	556
Q4	11.3	-0.383*	(0.198)	604
Q5	17.2	-0.385*	(0.219)	545
$T \times s - c $		-0.0253	(0.0176)	3154

Notes: Quintiles of absolute distance from the grade group assignment cutoff. OLS with strata FE and EB ability control. Standard errors are clustered at the school grade group level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 15: Effect on Within-Class Dispersion in Test Scores

	Baseline Scores			Predicted Outcomes			Endline Scores		
	Control mean (1)	Coef. (2)	N (3)	Control mean (4)	Coef. (5)	N (6)	Control mean (7)	Coef. (8)	N (9)
Panel A: Kenya									
Full Sample	0.663	-0.257*** (0.026)	7102	0.771	-0.267*** (0.021)	7102	0.661	-0.130*** (0.026)	4954
Lower Grade	0.648	-0.215*** (0.032)	3618	0.757	-0.222*** (0.027)	3618	0.659	-0.135*** (0.033)	2594
Upper Grade	0.678	-0.307*** (0.031)	3484	0.785	-0.321*** (0.027)	3484	0.663	-0.125*** (0.039)	2360
Panel B: Liberia									
Full Sample	0.542	-0.022 (0.029)	4784	0.669	-0.042 (0.034)	4784	0.448	0.092** (0.042)	3154
Lower Grade	0.529	0.029 (0.035)	2417	0.648	0.034 (0.040)	2417	0.421	0.059 (0.063)	1679
Upper Grade	0.554	-0.074 (0.046)	2367	0.689	-0.115** (0.054)	2367	0.481	0.135** (0.053)	1475

Notes: The outcome is the absolute deviation of each student's standardized score from the classroom mean. All specifications include randomization-strata fixed effects. The baseline score columns omit the baseline control because the outcome is constructed from baseline scores; the predicted-outcome and endline score columns control for standardized baseline score. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 16: Approximate Peer/Rank Diagnostic (Borusyak–Hull Control Function)

	Experiments	
	(1)	(2)
	Kenya	Liberia
Peer mean EB ability	-0.133** (0.052)	-0.048 (0.108)
N	4954	3153

Notes: Dependent variable is standardized endline score. Peer mean EB ability is the mean of classmates' EB ability, excluding the focal student. This is the approximate control function implementation: controls include baseline-decile $\times$ treatment $\times$ grade FE and strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 17: Kenya Peer/Rank Diagnostics: Approximate vs. Exact Borusyak–Hull Specifications

	(1)	(2)	(3)	(4)	(5)
	Decile FE	Ventile FE	Exact CF	Exact recentered	Exact CF (simple)
Peer mean EB ability	-0.133** (0.052)	-0.136** (0.053)	0.001 (0.085)	-0.009 (0.083)	-0.003 (0.075)
Exact μ_i^{BH} control			-0.177* (0.096)		-0.166* (0.088)
Own-score controls	Cell FE	Cell FE	Cubic EB $\times$ T $\times$ grade	Cubic EB $\times$ T $\times$ grade	Linear EB $\times$ T $\times$ grade
Strata FE	Yes	Yes	Yes	Yes	Yes
School-clustered SE	Yes	Yes	Yes	Yes	Yes
Observations	4954	4954	4952	4952	4952

Notes: Columns (1)–(2) report the approximate Kenya peer/rank diagnostic specifications using baseline score bins $\times$ treatment $\times$ grade fixed effects. Column (3) is the exact Borusyak–Hull control function specification: it regresses standardized endline score on the realized peer mean, excluding the focal student, and the exact expected peer regressor $\mu_i^{BH} = p_j P_i(1) + (1 - p_j) P_i(0)$, where $P_i(1)$ is the peer mean under the deterministic treated reading-group rule and $P_i(0)$ is the peer mean under grade-based control classrooms. Column (4) uses the exact recentered regressor $\tilde{P}_i = P_i^{obs} - \mu_i^{BH}$. Column (5) keeps the exact control function but replaces the flexible cubic own-score controls with a simpler linear EB $\times$ treatment $\times$ grade control set. All exact specifications include strata FE and clustered standard errors at the school level. The probability p_j is the constituency-specific assignment probability from the full Kenya year-1 randomization roster (50 of 292 schools assigned to treatment). ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 18: Peer/Rank Diagnostics and Accounting: Kenya

	Kenya specifications	
	(1)	(2)
	Decile FE	Ventile FE
<i>Panel A: Peer/rank composite ($\hat{\zeta}$)</i>		
Effect of 1 SD increase in peer mean	-0.133** (0.052)	-0.136** (0.053)
Observations	4954	4954
<i>Panel B: Treatment-induced classroom shifts</i>		
Δ mismatch, $ \hat{\theta} - \hat{I} $ (C – T)		0.421
Δ peer composition (T – C)		0.277
Δ class size (T – C)		-3.1
Δ within class baseline SD (C – T)		0.326
<i>Panel C: Accounting decomposition</i>		
Overall ITT	0.031 (0.052)	
Peer/rank contribution ($\hat{\zeta} \times \Delta \bar{\theta}$)	-0.037 (0.019)	
Remainder	0.067 (0.053)	

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient from regressing standardized endline scores on peer mean ability, excluding the focal student, conditioning on baseline score bin $\times$ treatment $\times$ grade fixed effects and strata FE. Standard errors are clustered at the school level. Panel B reports mean treatment control differences in classroom characteristics. Mismatch is measured as $|\hat{\theta}_i^{EB} - \hat{I}_k|$ where $\hat{I}_k$ is the population-level track mean. Panel C reports the implied peer/rank contribution under the approximate specification ($\hat{\zeta} \times$ observed peer shift) and a remainder that includes assignment gains, productive-time effects, and other non-peer/rank channels. Bootstrap SEs (1,000 replications, clustered at the school level) in parentheses in Panel C.

Table 19: Treatment Effects on Lesson Completion Rate (Kenya)

	Lesson completion rate			
	(1)	(2)	(3)	(4)
Treat	0.018 [0.014]	0.022 [0.015]	0.018 [0.016]	0.024 [0.017]
Teacher attendance		0.085*** [0.029]		0.090*** [0.034]
Class size			-0.001 [0.001]	-0.001 [0.001]
Control mean	0.172	0.172	0.173	0.173
Observations	1128	1122	780	776

Notes: Each column reports the coefficient from a regression of lesson completion rate on treatment status. The unit of observation is a school–subject–grade cell. Lesson completion is the fraction of scripted literacy-revision lessons completed, recorded in NewGlobe’s classroom management system. Columns 2–4 add controls for teacher attendance rate and reading class size. Control means are computed in each column’s estimation sample. Standard errors are clustered at the school level in brackets. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 20: Approximate Peer/Rank Diagnostic by Score Density

	Score-density tercile		
	(1) Low (tails)	(2) Medium	(3) High (middle)
<i>Panel A: Approximate peer/rank estimate by density tercile</i>			
Peer mean EB ability ($\hat{\zeta}$)	-0.179** (0.081)	-0.099* (0.053)	-0.123** (0.062)
N	1708	1672	1574
	Pooled interaction		
	<i>Panel B: Interaction specification</i>		
	Peer mean EB ability ($\hat{\zeta}$)	-0.141	
	Peer EB $\times$ density interaction	-0.024 (0.039)	

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient ($\hat{\zeta}$) separately for terciles of the baseline score density evaluated at each student’s own score. Students in the tails of the distribution (low density) have few classmates at similar ability levels, so their within class rank is insensitive to changes in peer composition. Students in the middle (high density) have many nearby classmates, so rank shifts more with peer changes. In the stylized rank model, $\hat{\zeta} \approx \pi + \eta \cdot (-f(\theta_i))$, where π is the mean peer effect, η is the rank effect, and $f(\theta_i)$ is the score density. If rank effects matter in this reduced-form composite ($\eta \neq 0$), $\hat{\zeta}$ would tend to be more negative in the high-density tercile, where rank is more responsive to peer shifts. Panel B reports a pooled interaction of peer mean with standardized density; the interaction is a diagnostic for whether the peer/rank coefficient changes with local score density. All specifications include baseline-decile $\times$ treatment $\times$ grade FE and strata FE, so they should be read as approximate cell-control specifications rather than exact design-based recentering checks. Density is estimated using a Gaussian kernel with Silverman bandwidth, computed within grade, and standardized to mean zero and unit variance within grade. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 21: Peer/Rank Diagnostics and Classroom Shifts: Liberia

	Liberia
	(1)
	Estimate
<i>Panel A: Approximate Borusyak–Hull peer/rank coefficient</i>	
$\hat{\zeta}_{Lib}$	-0.048 (0.108)
Observations	3153
<i>Panel B: Treatment-induced classroom shifts</i>	
Δ absolute EB-to-target mismatch (C – T)	0.348
Δ peer composition (T – C)	0.041
Δ class size (T – C)	11.4
Δ within class baseline SD (C – T)	0.058
Overall ITT	-0.211 (0.135)

Notes: Panel A reports the approximate Borusyak–Hull control function peer/rank coefficient using decile $\times$ treatment $\times$ grade FE. Standard errors are clustered at the school grade group level. Panel B reports treatment control differences in classroom characteristics. The near-zero peer/rank coefficient and small classroom composition shift provide little evidence that the measured peer/rank channel explains the negative Liberia ITT pattern.

Table 22: ITT Estimates under Alternative Specifications

	Kenya		Liberia	
	Coef.	SE	Coef.	SE
Baseline (EB + strata FE)	0.031	(0.053)	-0.212	(0.135)
Raw baseline score control	0.031	(0.053)	-0.212	(0.135)
No baseline control	0.110	(0.071)	-0.240*	(0.139)
Wild cluster bootstrap p value	0.622		0.126	
Permutation p value	0.573		0.128	
N	4954		3154	

Notes: Each row reports the ITT estimate of treatment on standardized endline scores under a different specification. The baseline controls for empirical Bayes ability and strata fixed effects, with standard errors clustered at the school level in Kenya and at the school grade group level in Liberia. The raw baseline score row replaces the EB shrinkage estimate with the standardized raw baseline score. Wild cluster bootstrap p values use 999 replications (Cameron et al. 2008). Permutation p values are from a Fisher-exact randomization test with 999 permutations within strata at the country-specific assignment unit: school in Kenya and school grade group in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 23: Lee (2009) Bounds on ITT Estimates

	Kenya	Liberia
Control attrition rate	0.314	0.341
Treatment attrition rate	0.229	0.340
Attrition diff. (T – C)	-0.085	-0.001
OLS ITT	0.031	-0.212
Lee lower bound	-0.023	-0.212
Lee upper bound	0.154	-0.212

Notes: Attrition is defined as having a baseline score but no endline score. Lee (2009) bounds trim the distribution of endline scores in the arm with lower attrition to equalize attrition rates, producing worst-case and best-case ITT estimates. All specifications control for EB ability and strata FE. The table reports point-estimate bounds; standard errors are not shown.

Table 24: Sorted Group Average Treatment Effects (GATES)

	Experiments	
	(1) Kenya	(2) Liberia
Group 1 (Q1 of $\hat{\tau}$)	0.238 (0.207)	-0.449** (0.220)
Group 2 (Q2 of $\hat{\tau}$)	0.090 (0.080)	0.163 (0.237)
Group 3 (Q3 of $\hat{\tau}$)	0.001 (0.062)	-0.066 (0.231)
Group 4 (Q4 of $\hat{\tau}$)	0.018 (0.060)	-0.036 (0.214)
Group 5 (Q5 of $\hat{\tau}$)	-0.180** (0.085)	-0.743*** (0.194)
Group 5 – Group 1	-0.418* (0.241)	-0.294 (0.287)
BLP heterogeneity p value	0.13	0.13

Notes: Each column reports the average treatment effect within quintiles of the machine-learning-predicted conditional average treatment effect $\hat{\tau}(X_i)$, estimated using a cluster-cross-fitted causal forest (Wager and Athey, 2018). Group 1 contains students with the lowest (most negative) predicted effects; Group 5 contains those with the highest. Baseline covariates used for prediction include baseline score, EB ability, grade, upper-track indicator, distance from track cutoff, and additional pre-treatment historical controls from baseline/historical records in the analysis files. The forest is supplied with known assignment probabilities from the analysis data: constituency probabilities in Kenya and randomization stratum probabilities in Liberia. The BLP heterogeneity p value tests the null that treatment effects are constant (Chernozhukov et al., 2025). Within-group estimates come from a single interacted regression that controls for baseline score and randomization-strata fixed effects, with standard errors clustered at the school level in Kenya and at the school grade group level in Liberia. This outcome-regression adjustment is separate from the richer forest prediction feature set above and uses the raw baseline score rather than the EB transform. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 25: Treatment Effects With and Without Class-Size Controls

	Kenya		Liberia	
	(1) No CS	(2) With CS	(3) No CS	(4) With CS
Lower track (Treatment)	0.046 (0.137)	0.043 (0.138)	-0.019 (0.181)	-0.108 (0.180)
T $\times$ Upper	-0.023 (0.147)	-0.023 (0.148)	-0.328 (0.201)	-0.237 (0.189)
Upper track total	0.023 (0.044)	0.021 (0.044)	-0.348** (0.154)	-0.345** (0.151)
Class size		-0.001 (0.002)		0.003 (0.002)
N	4954	4954	3154	3154

Notes: Odd columns reproduce the track-interaction specification without class size controls. Even columns add realized reading class size as a control. Upper track total is the sum of the Treatment and T $\times$ Upper coefficients. All columns control for EB ability and strata FE. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 26: Robustness to Score Ceiling Effects (Kenya)

	(1) OLS	(2) Tobit	(3) Trimmed OLS
Treatment	0.031 (0.053)	0.040 (0.058)	0.022 (0.057)
N	4954	4954	4471
Right-censored obs.		176	

Notes: Column (1) reproduces the baseline OLS ITT on standardized endline scores. Column (2) estimates a Tobit model on the raw endline composite (upper limit = 50 points), controlling for raw baseline score and strata FE; the coefficient and SE are divided by the control-group SD of raw endline scores for comparability with column (1). Column (3) re-estimates OLS after dropping students in the top baseline score decile. Columns (1) and (3) control for EB ability and strata FE; column (2) controls for raw baseline score and strata FE. Standard errors are clustered at the school level. ***, **, and * indicate significance at 1%, 5%, and 10%.

Table 27: Treatment Effect on Score Variance

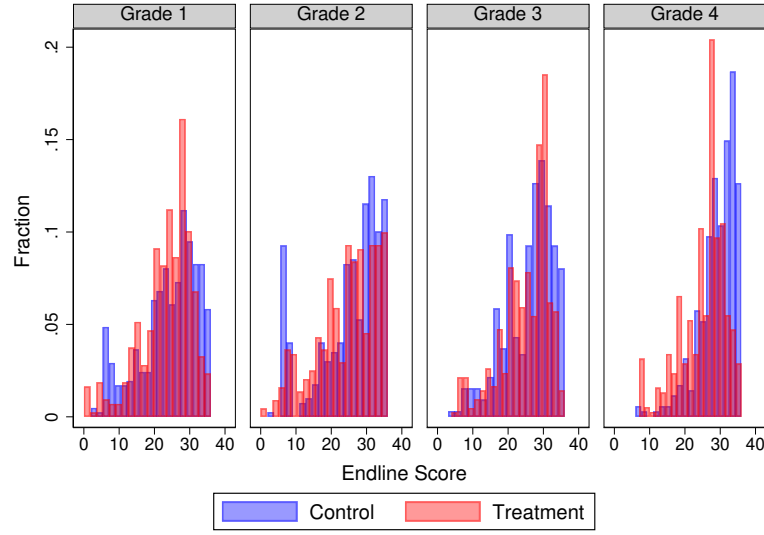
	Kenya	Liberia
<i>Panel A: Aggregate variance</i>		
Control	0.988	0.921
Treatment	0.915	0.997
<i>Panel B: Treatment on total variance</i>		
Treatment	0.038 (0.073)	0.054 (0.210)
<i>Panel C: Treatment on within class variance</i>		
Treatment	-0.228*** (0.052)	0.137* (0.069)
<i>Panel D: Treatment on between class variance</i>		
Treatment	0.239*** (0.050)	-0.082 (0.199)
N	4954	3154

Notes: Panel A reports raw variance of standardized endline scores by treatment arm. Panels B–D regress individual-level squared deviations on treatment, controlling for EB ability and strata FE. Total variance uses deviations from the sample grand mean; within class variance uses deviations from realized reading classroom means; between class variance uses the squared deviation of the classroom mean from the grand mean. Standard errors are clustered at the school level in Kenya and at the school grade group level in Liberia. ***, **, and * indicate significance at 1%, 5%, and 10%.

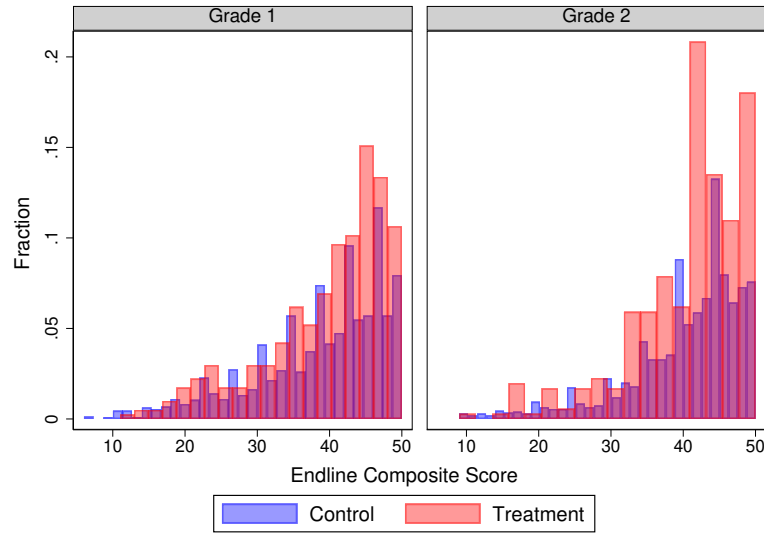
A Supplementary Descriptives and Diagnostics

Figures A1, A2, A3, A4, A5, and A6 collect supporting diagnostics for the main results. The endline and class size distributions show the realized outcome and classroom environments behind the reduced-form estimates. The Liberia model-implied misassignment and track-bin figures probe whether negative upper-track estimates are concentrated near the assignment margin. The sorted-GATES and variable-importance figures summarize the data-driven heterogeneity exercise reported in Table 24.

Figure A1: Endline Score Distributions by Grade and Treatment Status



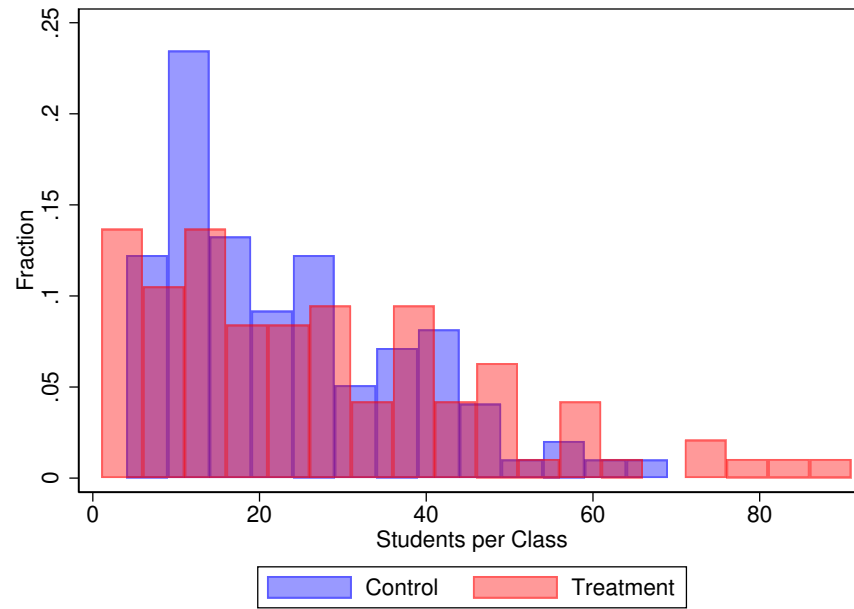
(a) Liberia



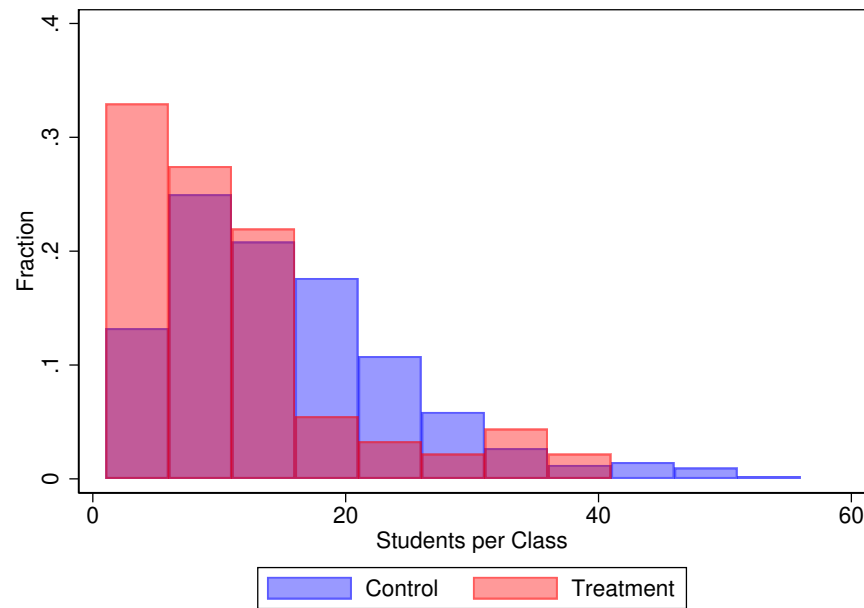
(b) Kenya

Notes: This figure plots kernel density estimates of endline reading scores for students with observed endline outcomes, separately by study, enrolled grade, and treatment assignment. Scores are standardized within grade to have mean zero and unit variance in the control group. The figure is descriptive and shows how the distribution of realized achievement shifts under treatment in each setting.

Figure A2: Reading-Class Size Distributions by Treatment Status



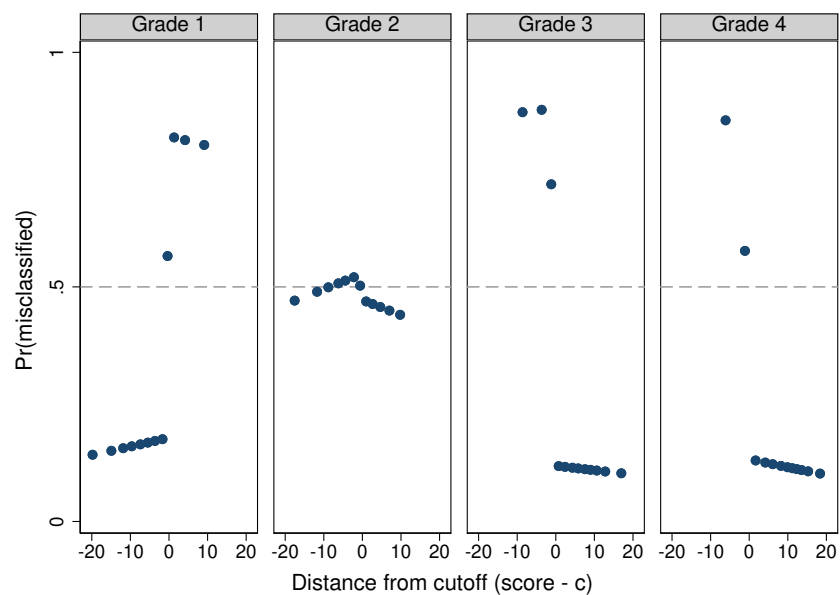
(a) Liberia



(b) Kenya

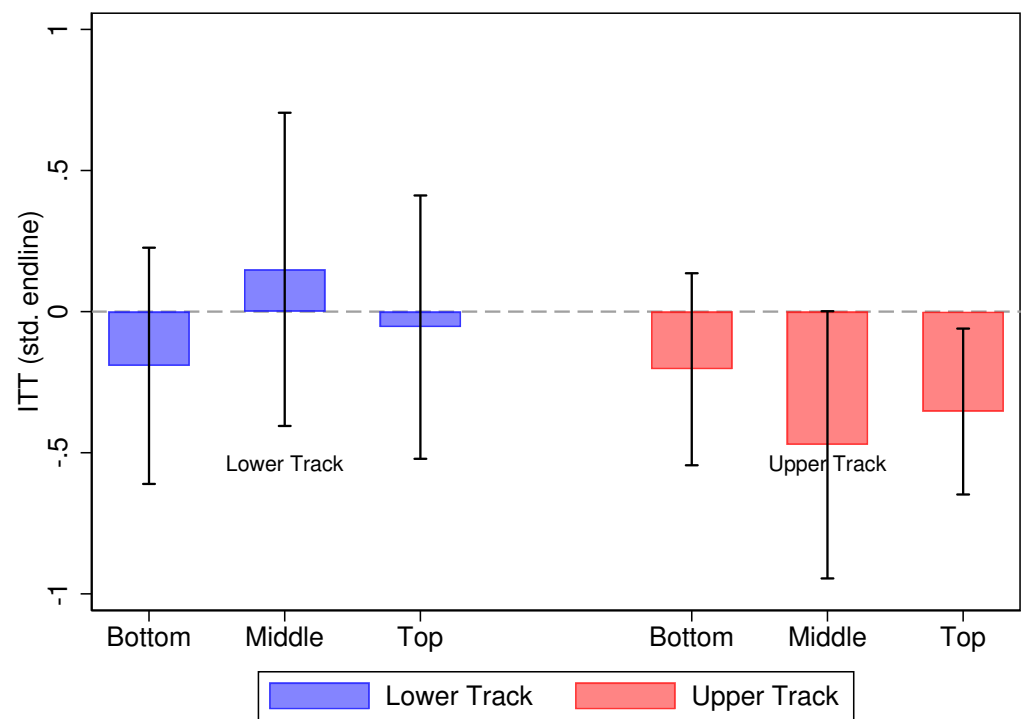
Notes: Each panel shows realized reading class size by treatment status; the unit is a reading classroom. Control units retain grade classrooms, while treatment units use cross-grade tracks. Liberia treatment classrooms are much larger; Kenya differences partly reflect baseline imbalance.

Figure A3: Model-Implied Latent Misassignment by Distance from the Assignment Cutoff in Liberia



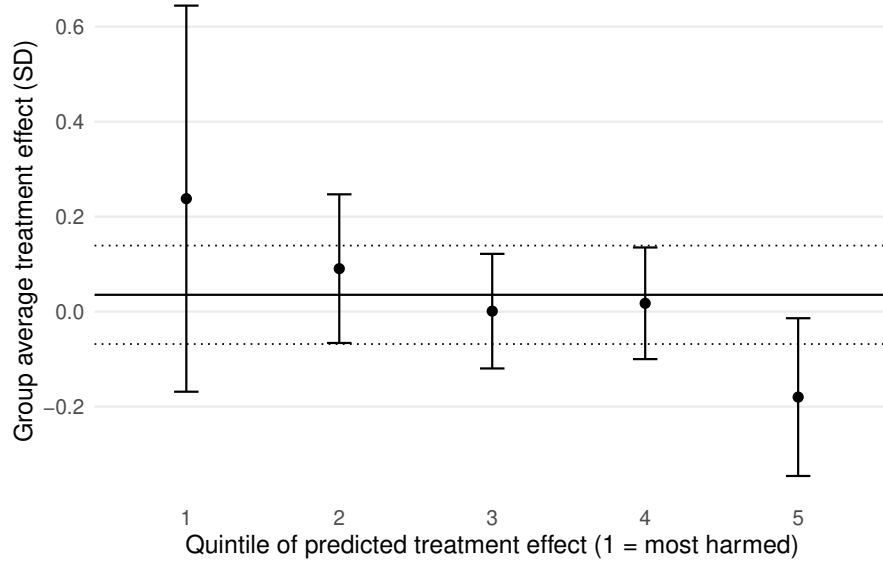
Notes: This figure plots the model-implied probability that a student's observed track assignment differs from the assignment that would obtain under latent ability, as a function of absolute distance from the Liberia grade group score cutoff. The probabilities are constructed from the Liberia empirical Bayes measurement system rather than from observed assignment errors, which are zero in the data. Under Liberia's low baseline predictive power, predicted latent misassignment remains substantial even for students away from the cutoff.

Figure A4: Liberia Treatment Effects by Track and Within-Track Ability Tercile

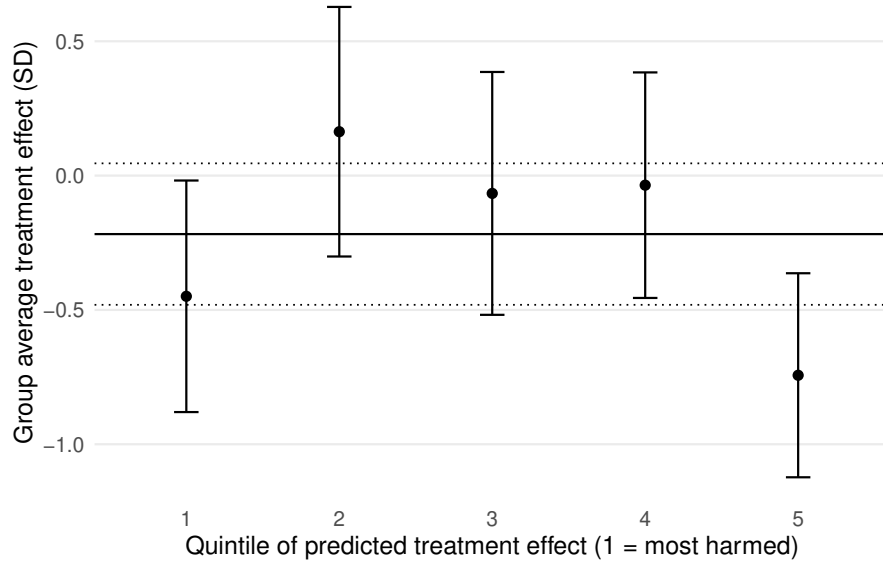


Notes: Bars report Liberia student-level intent-to-treat estimates for standardized endline reading scores, separately by assigned track and by tercile of baseline ability within that track. Error bars denote 95% confidence intervals from the corresponding OLS specifications with strata fixed effects, the empirical Bayes baseline control, and standard errors clustered at the school grade group level. Red bars indicate estimates with nominal $p < 0.05$ using the same clustered inference. The largest negative point estimates occur in the middle and top terciles of the upper track rather than among students nearest the lower-track margin.

Figure A5: Sorted GATES by Experiment



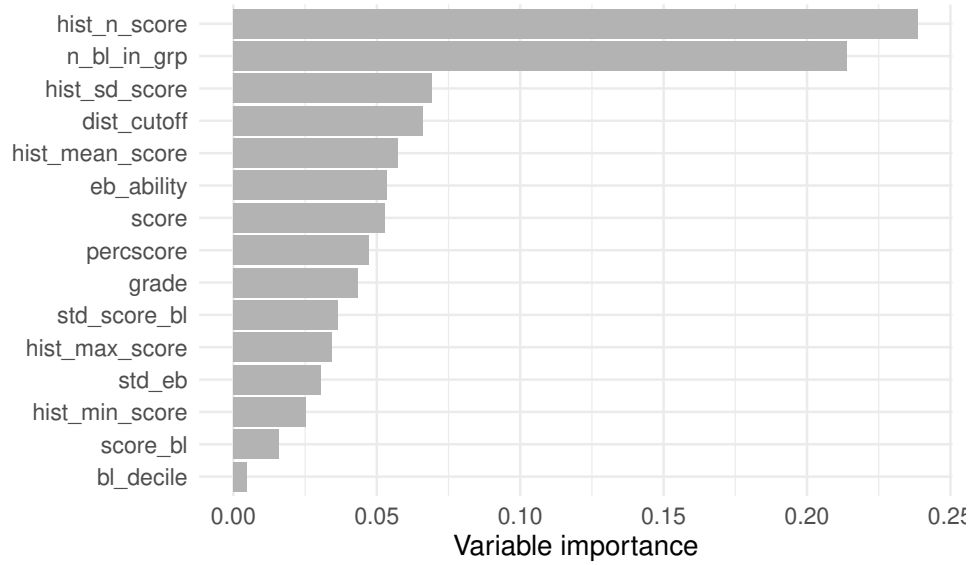
(a) Kenya



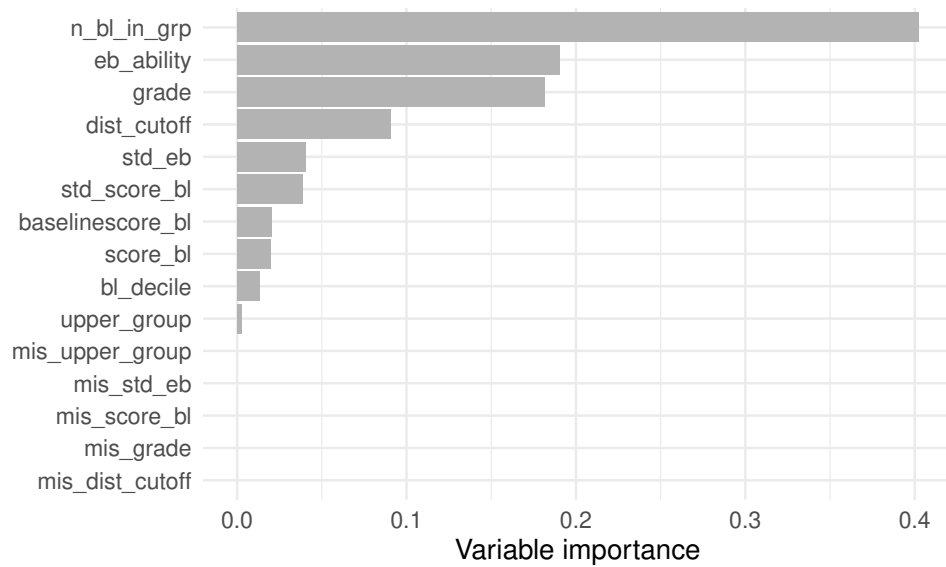
(b) Liberia

Notes: Dots report within-quintile treatment effects from the sorted GATES exercise, where students are grouped by quintiles of cluster-cross-fitted machine-learning-predicted treatment effects from a causal forest. Vertical bars denote 95% confidence intervals based on cluster-robust standard errors, clustered at the school level in Kenya and at the school grade group level in Liberia. The solid horizontal line marks the full-sample ITT estimate, and dotted horizontal lines mark its 95% confidence interval. Quintile 1 is the most negatively predicted group, and quintile 5 is the most positively predicted group.

Figure A6: Causal-Forest Variable Importance



(a) Kenya



(b) Liberia

Notes: Panels report variable-importance rankings from the causal forests used for sorted GATES. Higher values indicate greater contribution of a baseline covariate to treatment-effect prediction. Importance is descriptive and does not imply causal attribution.

B Sample Flow

Table A1: Sample Flow — Liberia

	Experiment (Liberia)		
	(1)	(2)	(3)
Sample flow step	Control	Treatment	Total
All observations	3420	4165	7585
Analytic sample	2249	2535	4784
With baseline score	2249	2535	4784
With endline score	1482	1672	3154

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.

Table A2: Sample Flow — Kenya

	Experiment (Kenya)		
	(1)	(2)	(3)
Sample flow step	Control	Treatment	Total
All observations	8271	1447	9718
Analytic sample	6117	985	7102
With baseline score	6117	985	7102
With endline score	4195	759	4954

Notes: Counts shown by treatment status at each sample construction step in the cleaned analysis dataset. The analytic sample keeps students with a nonmissing baseline score and at least two baseline scored students in the assignment unit used to form peer and classroom measures.